# PSDA FINAL EXAM WAR BOOK

Probability & Statistical Data Analysis — SCST1223

## From Zero to Ready in One Day

Built from 11 real past final exams (2012–2024) + the 2026 lecture slides

**Exam:** Thursday 16 July 2026, 2:30 PM

**Scope:** Chapter 5 (Estimation + Hypothesis Testing), Chapter 6 (Chi-Square), Chapter 7 (Correlation & Regression), Chapter 8 (ANOVA)

**Prepared for:** Al-Sakkaf — 15 July study day, Johor prayer times included

# PART 0 — READ THIS FIRST (20 minutes)

## 0.1 The truth about this exam (good news)

You start from zero. That is fine, because this exam does not test cleverness. It tests whether you can run the same recipe again and again with different numbers. After analysing every question in 11 past final exams, here is the truth:

- **Around 80% of the whole paper is ONE skill** — the 5-step hypothesis test — wearing different costumes (t-test, chi-square, F-test, ANOVA...). Learn the 5 steps once, then just learn which formula each costume uses.
- The other 20% is plug-into-formula arithmetic (confidence intervals, correlation r, regression line). No thinking, just careful calculator work.
- **The lecturers recycle questions.** Several questions appear in multiple years with identical or nearly identical numbers (proof in Part 0.3). Your practice questions in this book ARE past exam questions — so when you sit the real paper, most of it will feel familiar.

**Every question gives marks for steps, not only the final answer.** Even if your arithmetic slips, writing the correct hypotheses, formula, and decision rule collects most marks. The templates in this book are exactly those steps.

**Rule that applies to the whole paper: give final answers to 3 decimal places. The formula list and the statistical tables (z, t, chi-square, F) are PRINTED in the exam paper — you memorise almost nothing.**

## 0.2 The exam map — what the 6 questions always look like

Recent papers (2021/22, 2022/23, 2023/24) use exactly this shape. Older papers use the same topics in slightly different order. 100 marks total.

| Question | Marks | Topic (lesson in this book) | What you must do |
|---|---|---|---|
| Q1 | 8–13 | Point estimate + Confidence interval (L1), often + a small 1-sample test (L2/L3) | Compute mean or proportion, build CI, interpret it |
| Q2 | 15–22 | Two-sample tests (L4) + Paired t-test (L5) | Full 5-step hypothesis tests |
| Q3 | 20–25 | Chi-square (L6): goodness-of-fit + independence | Expected counts, chi-square statistic, decision |
| Q4 | 15–25 | Correlation (L7): scatter plot + Pearson/Spearman r | Sketch, compute r, interpret |
| Q5 | 10–20 | Regression (L8): least squares line + prediction + slope test | b1, b0, equation, predict, test slope |
| Q6 | 10–20 | ANOVA (L9): compare 3–4 group means | 8-step F test |

**Timing in the exam hall:** the paper is usually 3 hours for 100 marks. Simple rule: about 1.5 minutes per mark, so a 20-mark question deserves about 30 minutes. Q1 is the warm-up — do it fast.

## 0.3 The Golden Discovery — they recycle questions

This is the highest-value page in the book. Compare these, found while analysing your folder:

| Recycled question | Appeared in | Notes |
|---|---|---|
| Engine volume vs mileage (regression) | 2016/17 Q6 • 2021/22 Q5a • 2023/24 Q5a | Same story, nearly same numbers, 3 times |

| | | |
|---|---|---|
| Patient age vs blood pressure + slope test (sb1 = 0.07) | 2017/18 Q5 • 2021/22 Q5b • 2023/24 Q5b | 2021/22 and 2023/24 numbers are IDENTICAL |
| Weight before/after exercise (paired t) | 2016/17 Q3b • 2021/22 Q2c | Identical 12 people, identical numbers |
| Memory drug / placebo / none (ANOVA) | 2016/17 Q7 • 2018/19 Q5 | Identical numbers |
| President gender gap (chi-square independence) | 2013/14 Q4 • 2021/22 Q3b | Identical table: 325/151/324, 258/241/251 |
| Compare given chi-square vs critical values (conceptual) | 2013/14 Q4i • 2021/22 Q3a | Same 3 sub-questions, tiny number changes |
| Bacteria/mice lifespan (1-sample t) | 2016/17 Q2a • 2021/22 Q1b | Same skeleton, new numbers |
| Equal distribution across days (chi-square GOF) | 2014/15 • 2015/16 • 2022/23 • 2023/24 | Accidents / cars / reservations — every era |

**Pattern:** 2021/22 was largely a remix of 2013/14 + 2016/17. 2023/24 was largely a remix of 2021/22. So your exam will very likely remix the recent papers — and every one of those questions is a worked practice inside this book. Level 2 practice = literally past exam questions.

## 0.4 Words you must understand (your zero-to-one dictionary)

Only these few words stand between you and every question. Read slowly once; everything later uses them.

| Word | Meaning (simple) | Arabic anchor |
|---|---|---|
| Population | ALL the people/things we care about (all students, all bottles…) | المجتمع كله |
| Sample | The small group we actually measured (30 bottles from the factory) | العينة |
| Mean $\bar{x}$ or $\mu$ | The average. $\bar{x}$ = average of the sample; $\mu$ = average of the whole population | المتوسط |
| Standard deviation s or σ | How spread out the numbers are. Small = numbers close together | الانحراف المعياري |
| Variance $s^2$ or $\sigma^2$ | Standard deviation squared. Just (spread)² | التباين |
| Proportion $\hat{p}$ or p | A percentage as a decimal: 130 out of 350 shops → $\hat{p}$ = 0.371 | النسبة |
| Hypothesis | A claim about the population that we test using the sample | الفرضية |
| Significance level α | How much risk of a wrong 'reject' we accept: 0.05 = 5% | مستوى الدلالة |
| Reject H0 | The evidence is strong: the claim of 'no difference' falls | نرفض فرضية العدم |
| Fail to reject H0 | Evidence NOT strong enough. (We never say 'accept H0') | لا نرفضها |
| Degrees of freedom (df) | A number that tells you WHICH ROW of the table to read. Each test has a small df rule | درجات الحرية |
| Critical value | The border number from the table. Cross the border → reject H0 | القيمة الحرجة |

## 0.5 The Universal 5-Step Template (the heart of the whole exam)

This is the template your lecturer uses in the tutorial answer files, so write your answers exactly like this. Roughly **70–80 marks of the 100** are earned by running these 5 steps with the right formula.

**TEMPLATE — EVERY hypothesis test, always the same 5 steps**

**STEP 1 — Hypotheses.** Write H0: parameter = value (always =). Write H1 with ≠, < or > (copy the direction from the question words). Then say: two-tailed (≠), left-tailed (<) or right-tailed (>).

$$H_0:\ \mu = \mu_0, H_1:\ \mu \quad \neq\ /\ <\ /\ > \mu_0$$

**STEP 2 — Test statistic.** Pick the ONE correct formula (the chooser table in 0.6 tells you which). Substitute the numbers. Compute to 3 decimal places.

**STEP 3 — Critical value.** Use α and df to read the border number from the printed table. Draw a small bell curve, mark the critical value(s), shade the rejection tail(s).

**STEP 4 — Decision.** If the test statistic falls in the shaded region → Reject H0. Otherwise → Fail to reject H0.

**STEP 5 — Conclusion sentence.** One sentence, always this shape: 'At the α = 0.05 significance level, there is (sufficient / not sufficient) evidence to conclude that ...' and finish with the claim in the question's own words.

**How to phrase Step 5 correctly (this is worth marks):**

- **If the claim is in H1** (most questions): Reject H0 → 'there is sufficient evidence to SUPPORT the claim...'. Fail to reject → 'there is not sufficient evidence to support the claim...'.
- **If the claim is in H0** (words like 'is equal to', 'is 35 hours'): Reject H0 → 'there is sufficient evidence to REJECT the claim...'. Fail to reject → 'there is not sufficient evidence to reject the claim...'.

**Direction words → H1 symbol:** 'more than / greater / increased / higher / effective / longer' → > • 'less than / decreased / lower / shorter / cheating' → < • 'different / changed / not equal / relationship exists' → ≠

## 0.6 The Chooser — read the question, pick the weapon

On the exam, identifying WHICH test is half the battle. Ask yourself these questions in order:

| If the question says... | Test / tool | Formula key (on your formula sheet) | df |
|---|---|---|---|
| Estimate / confidence interval for a mean or proportion | CI (Lesson 1) | $\bar{x} \pm t\frac{s}{\sqrt{n}}$ / $\hat{p} \pm z\sqrt{\frac{\hat{p}}{}}$ | t: n−1 |
| ONE sample, test the mean, σ (population sd) known OR n > 30 | z-test (L2) | $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ | — |
| ONE sample, test the mean, σ unknown AND n ≤ 30 | t-test (L2) | $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ | n−1 |
| ONE sample, test variance or standard deviation | chi-square (L3) | $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ | n−1 |
| TWO independent groups, compare means, 'assume variances equal' | pooled t (L4) | t with $sp^2$ formula | n1+n2−2 |
| TWO independent groups, 'variances | Welch T (L4) | T with the big v formula | v (round |

| | | | |
|---|---|---|---|
| NOT equal' | | | DOWN) |
| TWO groups, compare proportions / percentages | 2-proportion z (L4) | z with pooled $\bar{p}$ | — |
| TWO groups, compare variability / consistency / spread | F-test (L4) | $F = \frac{s_1^2}{s_2^2}$ | (n1−1, n2−1) |
| SAME people measured before & after | paired t (L5) | $t = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}$ | n−1 |
| Counts in categories vs 'all equal' or given % | $\chi^2$ goodness-of-fit (L6) | $\chi^2 = \Sigma \frac{(o-e)^2}{e}$ | k−1 |
| A table of counts: is variable A related to variable B? | $\chi^2$ independence (L6) | $e = \frac{row \times col}{total}$ | (r−1)(c−1) |
| How strong is the straight-line relationship? | Correlation r (L7) | Pearson r (or Spearman for ranks) | t for r: n−2 |
| Find the line / predict y from x | Regression (L8) | b1, b0 formulas | slope t: n−2 |
| Compare the means of 3 OR MORE groups | ANOVA F (L9) | $F = \frac{n s_{\bar{x}}^2}{s_p^2}$ | (k−1, k(n−1)) |

## 0.7 How to read the printed tables (5 minutes each, with examples)

**The tables are attached to the exam paper. You only need to know where to put your finger.**

### A. The z table (normal distribution)

The printed table gives the area between 0 and z. You will mostly need these four numbers — memorise them and you barely touch the table:

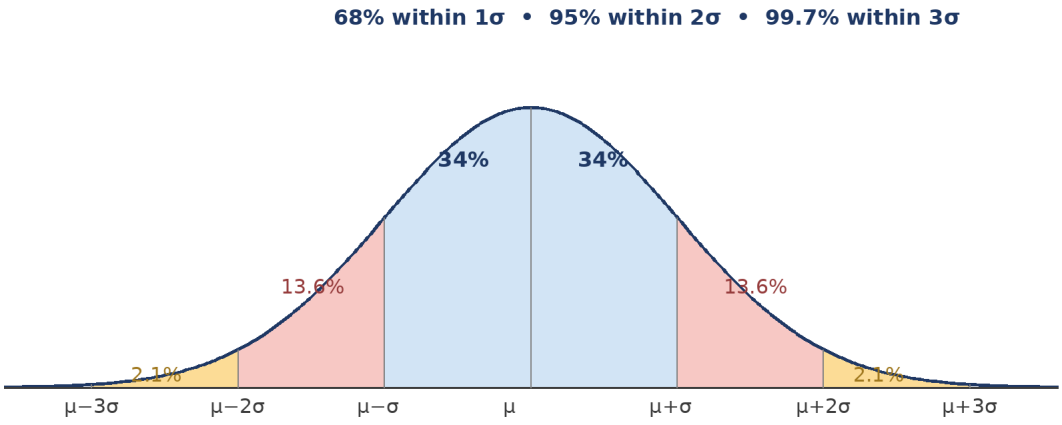



FIGURE 0.1 The normal curve: 68% within 1σ, 95% within 2σ, 99.7% within 3σ — the shape behind every z value.

| Use | α = 0.10 | α = 0.05 | α = 0.025 | α = 0.01 | α = 0.005 |
|---|---|---|---|---|---|
| z critical (one tail) | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 |
| Used for | 90% one-tail | 95% CI is ±1.960 \| one-tail 0.05 | 95% two-tail | 98% CI / one-tail 0.01 | 99% two-tail |

- Two-tailed test at α = 0.05 → split α in half → use z(0.025) = ±1.960.
- One-tailed test at α = 0.05 → use 1.645 (negative −1.645 if left-tailed).
- Confidence interval: 90% → 1.645, 95% → 1.960, 99% → 2.576.

### B. The t table

- Rows = degrees of freedom (df). Columns = α in ONE tail.
- One-tailed α = 0.05, df = 9 → column 0.05, row 9 → 1.833.
- **Two-tailed α = 0.05 → use column 0.025** (half of 0.05), e.g. df = 9 → 2.262. Halving α for two tails is the #1 table mistake.

### C. The chi-square ($\chi^2$) table

- Rows = df. Columns = area to the RIGHT of the value.
- Right-tail test, α = 0.05, df = 4 → column 0.05, row 4 → 9.488.
- **Two-tailed variance test needs BOTH borders:** left border = column (1 − α/2) e.g. 0.975; right border = column α/2 e.g. 0.025. Example df = 29, α = 0.01 → 13.121 and 52.336.

### D. The F table

- There is a separate F table for each α (0.05, 0.025, 0.01). Numerator df runs across the top; denominator df runs down the side.
- Example: α = 0.05, numerator 2, denominator 12 → F = 3.89.

- In ANOVA: numerator = k − 1 (groups minus 1), denominator = k(n − 1). In the two-variance F-test: (n1 − 1, n2 − 1).

## 0.8 Calculator drill (do this before anything else, 10 minutes)

Two skills you will repeat 20+ times in the exam. Train them until they take 60 seconds:

- **Mean:** add all values, divide by n. For 50, 55, 48, 52, 49 → sum 254 → $\bar{x}$ = 50.8.
- **Sample standard deviation s:** subtract the mean from each value, square, add them all, divide by (n − 1), square root. For the 5 values above: deviations −0.8, 4.2, −2.8, 1.2, −1.8 → squares 0.64+17.64+7.84+1.44+3.24 = 30.8 → 30.8/4 = 7.7 → s = √7.7 = 2.775.
- **Best: use your calculator's statistics mode (SD mode).** Casio fx: MODE → STAT → 1-VAR, enter values, read $\bar{x}$ and sx directly. Practise once tonight — it will save you 30+ minutes in the exam. Make sure you read sx (sample sd, divides by n−1), NOT σx.

## 0.9 Your battle schedule — Wednesday 15 July (prayer-aware)

Prayer times for Johor Bahru (JAKIM zone JHR02, 15 Jul 2026): Subuh 5:52 • Zohor 1:13 • Asar 4:38 • Maghrib 7:19 • Isyak 8:33. Each masjid break below includes walking time. Total study ≈ 11.5 hours — enough to cover every lesson.

| Time | Block | What you do |
|---|---|---|
| 8:00 – 8:20 | Part 0 | Read Part 0 fully. Set up calculator STAT mode |
| 8:20 – 9:30 | Lesson 1 | Confidence intervals — read + do all 3 practice levels |
| 9:30 – 11:00 | Lesson 2 | The 5-step engine + 1-sample z & t tests |
| 11:00 – 11:15 | Break | Walk, water, no phone scrolling |
| 11:15 – 12:00 | Lesson 3 | Variance chi-square test |
| 12:00 – 1:00 | Lesson 4 (part 1) | Two-sample means: pooled t + Welch T |
| 1:00 – 2:00 | Zohor + lunch | Masjid + eat. Step fully away from books |
| 2:00 – 3:00 | Lesson 4 (part 2) | Two proportions + F-test + practice |
| 3:00 – 3:50 | Lesson 5 | Paired t-test (appears EVERY year) |
| 3:50 – 4:30 | Lesson 6 (part 1) | Chi-square goodness-of-fit |
| 4:30 – 5:25 | Asar | Masjid + short walk |
| 5:25 – 6:15 | Lesson 6 (part 2) | Chi-square independence |
| 6:15 – 7:10 | Lesson 7 | Correlation (Pearson + Spearman) |
| 7:10 – 8:10 | Maghrib + dinner | Masjid + eat |
| 8:10 – 8:30 | Lesson 8 (part 1) | Regression: b1, b0, equation |
| 8:30 – 9:20 | Isyak | Masjid |
| 9:20 – 10:00 | Lesson 8 (part 2) | Prediction, $R^2$, slope test, R output |
| 10:00 – 11:00 | Lesson 9 | ANOVA 8-step + practice. Then SLEEP — no all-nighter |

**Thursday 16 July (exam day):** 8:30–11:00 sit the Mock Final (Part 10) under exam conditions • 11:00–12:00 mark it with the answer key and revisit weak lessons • 12:00–12:45 re-read every TEMPLATE box + the Chooser • Zohor + lunch • arrive at the hall by 2:00 PM. Exam 2:30 PM.

**If you fall behind schedule:** never skip Lessons 4, 5, 6 or 9 (≈60 marks live there). Shorten Lesson 3 and Lesson 7's Level-3 practice first.

# LESSON 1 POINT ESTIMATE & CONFIDENCE INTERVALS

**On the final exam:** Question 1, worth 8–13 marks. Fast marks — aim for full marks here.

**Seen in:** Every single year: 2013/14, 2015/16, 2016/17, 2017/18, 2018/19, 2021/22, 2022/23, 2023/24

**Study time:** 70 min | **Level for a beginner:** Easy — pure formula work

## The idea in plain words

**Point estimate** = your single best guess of the population value, taken from the sample. Best guess of the population mean μ is simply the sample mean $\bar{x}$. Best guess of a population proportion p is the sample proportion $\hat{p}$ = (number of successes) ÷ (sample size). That is genuinely all.

**Confidence interval (CI, فترة الثقة)** = your guess ± a safety margin. Instead of saying 'the mean is 51.3', you say 'I am 90% confident the true mean is between 50.1 and 52.6'. The safety margin is called the margin of error (ME).

**Every CI has the same skeleton:** estimate ± (table number) × (standard error). Only the ingredients change.

## The three CI recipes (all on your formula sheet)

| Situation | Formula | Table number |
|---|---|---|
| Mean, σ known (or n > 30) | $\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$ | z: 90%→1.645, 95%→1.960, 99%→2.576 |
| Mean, σ unknown and n ≤ 30 | $\bar{x} \pm t\frac{s}{\sqrt{n}}$ | t table, df = n − 1, column = (1−conf)/2 |
| Proportion | $\hat{p} \pm z\sqrt{\frac{\hat{p}\hat{q}}{n}}, \quad \hat{q} = 1 - \hat{p}$ | same z numbers |

## TEMPLATE — Confidence interval (write these 4 lines)

**1.** Point estimate: $\bar{x}$ = … (or $\hat{p}$ = x/n = …)

**2.** Table value: 90% → z = 1.645 / 95% → z = 1.960 / t-case: df = n−1, read column 0.05 (90%) or 0.025 (95%)

**3.** Margin of error: ME = (table value) × s/√n (or √($\hat{p}\hat{q}$/n) for proportion). 3 decimals.

**4.** CI = estimate ± ME → (lower, upper). Then interpret: 'We are 95% confident that the true population mean (say what it is in the story) lies between … and …'.

## WORKED EXAMPLE — the real 2023/24 Q1(a): wheat yield

15 wheat fields give yields: 50, 55, 48, 52, 49, 56, 53, 47, 51, 50, 54, 52, 48, 50, 55. (i) Point estimate of μ. (ii) Build and interpret a 90% CI for μ.

**MODEL ANSWER — wheat yield**

**(i)** the point estimate of μ is the sample mean:

$$\bar{x} = \frac{\Sigma x}{n} = \frac{770}{15} = 51.333 \quad KG \ \ per \ \ acre$$

**(ii)** σ unknown, n = 15 ≤ 30 → use t. df = 15 − 1 = 14. 90% → α = 0.10 → column 0.05 → t = 1.761. Calculator: s = 2.820.

$$ME = 1.761 \times \frac{2.820}{\sqrt{15}} = 1.761 \times 0.728 = 1.282$$

$$CI = 51.333 \pm 1.282 = (50.051,\ 52.615)$$

Interpretation: We are 90% confident that the true mean wheat yield of all farms lies between 50.051 and 52.615 KG per acre.

## Bonus recipe — minimum sample size (asked 2017/18)

**'How many must we survey for margin of error E?' →** n = $\hat{p}\hat{q}(z/E)^2$. Always ROUND UP (people can't be fractions).

**TRAPS — where students lose marks**

Choosing z when the question wants t: if σ (population sd) is NOT given and n ≤ 30 → t. The word 'sample standard deviation' = s → t-case.

Forgetting to halve: 95% CI uses column 0.025 in the t table (not 0.05).

Proportion: n is the TOTAL surveyed, x is the count that said yes. $\hat{p}$ = x/n. Do not mix them.

Round only the FINAL numbers to 3 decimals; keep full digits in your calculator between steps.

Interpretation sentence is worth marks — never skip it.

## PRACTICE · LEVEL 1 (warm-up)

A sample of 36 batteries has mean life $\bar{x}$ = 80 hours. Population σ = 12 hours. Build a 95% CI for μ.

**MODEL ANSWER — Level 1**

σ known → z = 1.960.

$$ME = 1.960 \times \frac{12}{\sqrt{36}} = 1.960 \times 2 = 3.920$$

$$CI = 80 \pm 3.920 = (76.080,\ 83.920)$$

We are 95% confident the true mean battery life is between 76.080 and 83.920 hours.

## PRACTICE · LEVEL 2 (real exam: 2022/23 Q1a + 2016/17 Q1b)

(a) 158 out of 367 students actively use ChatGPT. (i) Point estimate of the true proportion. (ii) 95% CI, with interpretation.

(b) 126 police officers: mean blood lead level 29.2 μg/dl, σ = 7.5. Build a 90% CI for the mean.

**MODEL ANSWER — Level 2**

**(a)** $\hat{p}$ = 158/367 = 0.431, $\hat{q}$ = 0.569.

$$ME = 1.960 \times \sqrt{\frac{0.431 \times 0.569}{367}} = 1.960 \times 0.026 = 0.051$$

$$CI = 0.431 \pm 0.051 = (0.380,\ 0.481)$$

We are 95% confident that between 38.0% and 48.1% of all students actively use ChatGPT.

**(b)** σ known → z = 1.645.

$$ME = 1.645 \times \frac{7.5}{\sqrt{126}} = 1.645 \times 0.668 = 1.099$$

$$CI = 29.2 \pm 1.099 = (28.101,\ 30.299)$$

## PRACTICE · LEVEL 3 (monster: 2017/18 Q1 + concept questions)

(a) 130 of 350 food shops had special offers. (i) 95% CI for the proportion. (ii) What sample size is needed for a 95% CI with margin of error 0.04?

(b) Concept (2015/16 Q1): Two 95% CIs for mean car speed were (50.3, 52.7) and (49.8, 50.6). Same sample sd. Which came from the LARGER sample?

### MODEL ANSWER — Level 3

**(a-i)** $\hat{p}$ = 130/350 = 0.371, $\hat{q}$ = 0.629.

$$ME = 1.960 \times \sqrt{\frac{0.371 \times 0.629}{350}} = 0.051\ , \qquad CI = (0.320,\ 0.422)$$

**(a-ii)** Sample size formula, then round UP:

$$n = \hat{p}\hat{q}\left(\frac{z}{E}\right)^2 = 0.371 \times 0.629 \times \left(\frac{1.960}{0.04}\right)^2 = 0.2334 \times 2401 = 560.4 \rightarrow n = 561\ sh$$

**(b)** The second one, (49.8, 50.6): its width is 0.8 versus 2.4. Bigger n → smaller σ/√n → narrower interval. Width shrinks as sample grows.

# LESSON 2 THE 5-STEP ENGINE + ONE-SAMPLE z AND t TESTS

**On the final exam:** Question 1(b)/Question 2, 5–12 marks — AND the 5 steps you learn here power Lessons 3–9 (≈75 marks).

**Seen in:** Every year without exception

**Study time:** 90 min — the most important lesson in the book | **Level for a beginner:** Medium at first, automatic after 3 practices

### The idea in plain words

**The story is always the same.** Someone makes a claim about a population ('the machine fills 32 ml', 'mean return is more than 8%'). We can't measure everyone, so we take a sample. Then we ask ONE question: is the gap between the sample result and the claim big enough to matter, or is it just random noise?

**The test statistic is a 'how many standard errors away' score.** z = 2.66 means the sample sits 2.66 standard errors above the claim. Far from 0 = suspicious. The critical value from the table draws the border: beyond it → we reject the claim H0.

**H0 (فرضية العدم) always carries '='. H1 carries the direction you are trying to prove.**

## Choosing z or t (10-second rule)

| Given in question | Use | Formula | Critical value from |
|---|---|---|---|
| σ (population sd) known — or n > 30 | z-test | $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ | z table |
| only s (sample sd), and n ≤ 30 | t-test | $t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$ | t table, df = n − 1 |

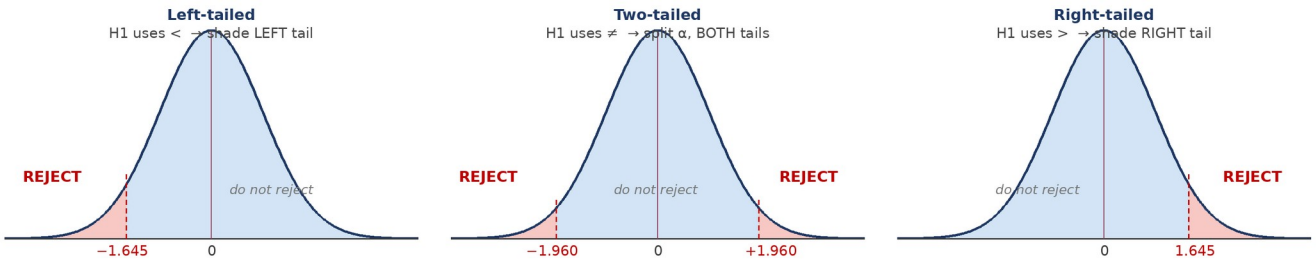



FIGURE 2.1 The three tail types. In Step 3, draw the matching curve and shade the rejection region.

**Always draw this little curve in Step 3 — the exams award marks for it ('draw and shade the critical region').**

## WORKED EXAMPLE A — the real 2023/24 Q1(b): stock returns

Analyst claims mean stock return is MORE than 8%. Sample: n = 60, $\bar{x}$ = 9.2%, σ = 3.5%. Test at α = 0.05.

### MODEL ANSWER — stock returns (z-test, right-tailed)

**Step 1:** 'greater than' → right-tailed test:

$$H_0: \mu = 8 \ , H_1: \mu > 8$$

**Step 2:** σ known → z-test:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{9.2 - 8}{3.5 / \sqrt{60}} = \frac{1.2}{0.452} = 2.656$$

**Step 3:** α = 0.05 right-tailed → critical z = 1.645. Draw bell, shade right of 1.645.

**Step 4:** 2.656 > 1.645 → test statistic is inside the rejection region → Reject H0.

**Step 5:** At the 0.05 significance level, there is sufficient evidence to support the claim that the mean stock return is greater than 8%.

## WORKED EXAMPLE B — the real 2021/22 Q1(b): bacteria

Claim: lifespan mean is LESS than 21 hours. Sample: n = 31 bacteria, $\bar{x}$ = 18, s = 4.8. α = 0.10.

### MODEL ANSWER — bacteria (t-test, left-tailed)

**Step 1:** left-tailed test:

$$H_0: \mu = 21 \ , H_1: \mu < 21$$

**Step 2:** σ unknown, n ≤ 30 → t-test:

$$t = \frac{18 - 21}{4.8 / \sqrt{31}} = \frac{-3}{0.862} = -3.480$$

**Step 3:** df = 30, one-tail α = 0.10 → t table → 1.310 → critical value is −1.310 (left tail). Draw and shade left.

**Step 4:** −3.480 < −1.310 → Reject H0.

**Step 5:** At the 0.10 significance level, there is sufficient evidence that the mean lifespan is less than 21 hours.

### TRAPS — where students lose marks

H0 ALWAYS gets '='. Writing H0: μ > 8 loses the Step-1 marks instantly.

The direction of H1 comes from the CLAIM WORDS, not from the data. 'Is it less than 21?' → H1: μ < 21 even before you see numbers.

**Negative test statistic with a left tail is normal.** Compare −3.480 with −1.310: reject because it is MORE negative (further left).

Two-tailed test: critical = ±value; reject if the statistic is beyond EITHER side.

Never write 'accept H0'. The professional wording is 'fail to reject H0'.

n > 30 lets you use z even when only s is given (e.g., n = 40, 50, 60...). n ≤ 30 with s → t.

## SIDE-SKILL Type I and Type II errors (asked as definitions in 2012/13 and 2014/15 — 4-5 free marks)

| Error | What it means | Memory hook |
|---|---|---|
| Type I (α) | Rejecting H0 when H0 is actually TRUE (a false alarm) | You 'convicted an innocent' claim. Probability = α |

| | | |
|---|---|---|
| Type II (β) | FAILING to reject H0 when H0 is actually FALSE (a miss) | You 'let the guilty go free' |

**Exam phrasing (2014/15):** internet provider claims speed is 400 kbps; customer suspects slower → H0: μ = 400, H1: μ < 400. Type I error = concluding the speed is slower than 400 when it really is 400. Type II error = concluding it is 400 (not rejecting) when it really is slower. Just translate the two definitions into the story's words.

## PRACTICE · LEVEL 1 (warm-up)

A factory claims the mean weight is 100 g. Sample: n = 25, $\bar{x}$ = 104, σ = 10. Test at α = 0.05 whether the mean is DIFFERENT from 100.

### MODEL ANSWER — Level 1

$$H_0:\ \mu = 100\ ,H_1:\ \mu \neq 100 \qquad (two\text{-}tailed)$$

$$z = \frac{104 - 100}{10/\sqrt{25}} = \frac{4}{2} = 2.000$$

Critical: two-tailed 0.05 → ±1.960. Since 2.000 > 1.960 → Reject H0.

At the 0.05 level, there is sufficient evidence that the mean weight is different from 100 g.

## PRACTICE · LEVEL 2 (real exam: 2018/19 Q1b + 2013/14 Q2i)

(a) Hinton Press claims press life is 14,500 hours, σ = 2,800. Sample of 25 gives $\bar{x}$ = 13,000. At α = 0.01, is the average life LESS than claimed?

(b) Bottles are labelled 32 ml. Auditor: n = 50, $\bar{x}$ = 31.8, s = 0.75, α = 0.01. Company is cheating if it fills less than 32 ml. Test it.

### MODEL ANSWER — Level 2

**(a)** left-tailed test:

$$H_0:\ \mu = 14500\ ,H_1:\ \mu < 14500$$

$$z = \frac{13000 - 14500}{2800/\sqrt{25}} = \frac{-1500}{560} = -2.679$$

Critical −z(0.01) = −2.326. Since −2.679 < −2.326 → Reject H0. Sufficient evidence the mean press life is less than 14,500 hours.

**(b)** n = 50 > 30 → z, left-tailed:

$$H_0:\ \mu = 32\ ,H_1:\ \mu < 32$$

$$z = \frac{31.8 - 32}{0.75/\sqrt{50}} = \frac{-0.2}{0.106} = -1.886$$

Critical −2.326. Since −1.886 is NOT beyond −2.326 → Fail to reject H0. At the 0.01 level there is not sufficient evidence that the company is cheating. (Yes — 'fail to reject' IS a legitimate exam answer.)

## PRACTICE · LEVEL 3 (monster: 2016/17 Q2b — raw data)

Claim: containers hold 10 litres on average. Contents of 10 random containers: 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8. Test at α = 0.01 (two-tailed), assuming normality.

**MODEL ANSWER — Level 3**

First compute from raw data (calculator STAT mode): $\bar{x}$ = 100.6/10 = 10.060, s = 0.246.

$$H_0:\ \mu = 10\ ,H_1:\ \mu \neq 10 \qquad (two\text{-}tailed)$$

n = 10 ≤ 30, σ unknown → t, df = 9.

$$t = \frac{10.060 - 10}{0.246/\sqrt{10}} = \frac{0.060}{0.078} = 0.772$$

Critical: two-tailed α = 0.01 → column 0.005, df 9 → ±3.250.

0.772 is between −3.250 and 3.250 → Fail to reject H0. At the 0.01 level, there is not sufficient evidence to reject the claim that the average content is 10 litres.

**What made this 'monster':** you had to build $\bar{x}$ and s yourself, halve α for two tails, and keep calm when the answer was 'fail to reject'.

# LESSON 3 TESTING A VARIANCE OR STANDARD DEVIATION ($\chi^2$ TEST)

**On the final exam:** Inside Q1/Q2 for 5–8 marks. Short, mechanical, easy to bank.

**Seen in:** 2013/14, 2014/15, 2017/18 (both semesters), 2018/19, 2022/23

**Study time:** 45 min | **Level for a beginner:** Easy once you respect the two-tailed table trick

## The idea in plain words

**Same 5-step engine, new costume.** Now the claim is about spread (variance $\sigma^2$ or standard deviation $\sigma$), not the mean. Example: 'the filling machine's variance is 35'. One formula does everything:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \quad with \quad df = n - 1$$

**Read the question carefully:** if it gives standard deviations, SQUARE them to get variances before substituting. $\sigma_0^2$ is the claimed value (from H0); $s^2$ is from your sample.

**The $\chi^2$ curve is not symmetric,** so a two-tailed test needs two DIFFERENT critical numbers from the table: left border = column $(1 - \alpha/2)$, right border = column $(\alpha/2)$.

## TEMPLATE — variance $\chi^2$ test

1. H0: $\sigma^2$ = value  H1: $\sigma^2$ ($\neq$ / < / >) value — square any sd the question gives.

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \rightarrow \quad 3 \ decimals$$

3. df = n − 1. Two-tailed: read columns $1-\alpha/2$ AND $\alpha/2$ (e.g. $\alpha = 0.01 \rightarrow$ columns 0.995 and 0.005). One-tailed right: column $\alpha$. One-tailed left: column $1-\alpha$.

4. Reject H0 if $\chi^2$ is beyond a border (bigger than the right border, or smaller than the left border).

5. Conclusion sentence with the claim in words.

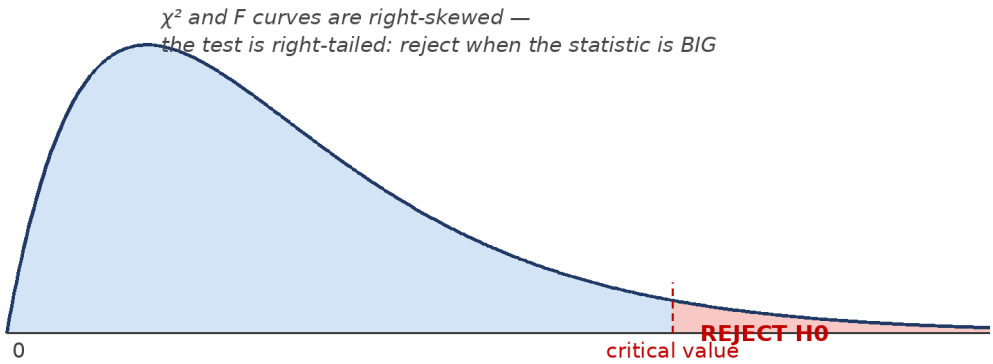



FIGURE 3.1 $\chi^2$ and F curves are right-skewed. Two-tailed variance tests read BOTH border columns.

## WORKED EXAMPLE — the real 2022/23 Q1(b): coffee time

30 staff, sample sd s = 6 hours. Claim: population variance is 35. Test at α = 0.01 (two-tailed: 'reject the claim that $\sigma^2 = 35$').

### MODEL ANSWER — coffee variance

**Step 1:** two-tailed:

$$H_0: \sigma^2 = 35 \ , H_1: \sigma^2 \neq 35$$

**Step 2:** substitute into the variance formula:

$$\chi^2 = \frac{(30 - 1) \times 6^2}{35} = \frac{29 \times 36}{35} = 29.829$$

**Step 3:** df = 29; α = 0.01 → left border = $\chi^2(0.995) = 13.121$, right border = $\chi^2(0.005) = 52.336$. Draw the skewed curve, shade both tails.

**Step 4:** 13.121 < 29.829 < 52.336 → inside the middle → Fail to reject H0.

**Step 5:** At the 0.01 level, there is not enough evidence to reject the claim that the population variance is 35 hours.

### TRAPS — where students lose marks

Given sd, claim about sd? SQUARE BOTH. 'sd hypothesised 50, sample sd 57, n = 30' → $\chi^2 = 29 \times 57^2/50^2$.

The table gives 'area to the right'. Left border for two-tailed α = 0.05 is column 0.975 — NOT 0.025.

This $\chi^2$ has df = n − 1 (sample size!). Don't confuse with Lesson 6's chi-square (categories).

One-tailed LEFT test ('variance decreased') → single border at column 1−α; reject if $\chi^2$ is SMALLER than it.

## PRACTICE · LEVEL 1 (warm-up, 2014/15 Q1b)

Machine 'out of control' if variance exceeds 1.15. Sample: n = 25, $s^2 = 2.03$. α = 0.05. Out of control?

### MODEL ANSWER — Level 1

$$H_0: \sigma^2 = 1.15 \ , H_1: \sigma^2 > 1.15$$

$$\chi^2 = \frac{24 \times 2.03}{1.15} = 42.365$$

Right-tailed: df = 24, column 0.05 → critical 36.415. Since 42.365 > 36.415 → Reject H0.

At the 0.05 level there is sufficient evidence that the machine is out of control.

## PRACTICE · LEVEL 2 (real exam: 2013/14 Q2ii)

25 computer users, s = 5 hours, claim $\sigma^2 = 35$, α = 0.05. Enough evidence to reject the claim?

### MODEL ANSWER — Level 2

$$H_0: \sigma^2 = 35 \ , H_1: \sigma^2 \neq 35 \ , \qquad df = 24$$

$$\chi^2 = \frac{24 \times 25}{35} = 17.143$$

Borders: $\chi^2(0.975) = 12.401$ and $\chi^2(0.025) = 39.364$. 17.143 sits between them → Fail to reject H0.

Not enough evidence to reject the claim that the variance is 35.

## PRACTICE · LEVEL 3 (monster: 2017/18 S2 Q2b + S1 Q2)

(a) Experienced inspectors: variance 0.18. New inspector: 101 stampings, $s^2 = 0.13$. Two-tailed, $\alpha = 0.05$: satisfactory measurements?

(b) Laundry machines: old $\sigma = 0.095$; after adjustment $n = 20$, $s = 0.065$. Did sd DECREASE? $\alpha = 0.05$.

### MODEL ANSWER — Level 3

**(a)** two-tailed:

$$H_0: \sigma^2 = 0.18 \;, H_1: \sigma^2 \neq 0.18$$

$$\chi^2 = \frac{100 \times 0.13}{0.18} = 72.222$$

df = 100 → borders $\chi^2(0.975) = 74.222$ and $\chi^2(0.025) = 129.561$. Since 72.222 < 74.222 → in the LEFT tail → Reject H0. The new inspector's variance is unusually small — measurements are NOT considered satisfactory (possibly discarding extreme readings).

**(b)** left-tailed; square the sd's:

$$H_0: \sigma = 0.095 \;, H_1: \sigma < 0.095$$

$$\chi^2 = \frac{19 \times 0.065^2}{0.095^2} = \frac{0.080275}{0.009025} = 8.895$$

Left critical: df = 19 column 0.95 → 10.117. Since 8.895 < 10.117 → Reject H0 → yes, the company successfully reduced the variability.

**Monster factor:** (a) rejects on the LEFT side of a two-tailed test — most students only check the right; (b) works with sd's that must be squared and a left-tail border.

# LESSON 4 TWO-SAMPLE TESTS (MEANS, PROPORTIONS, VARIANCES)

**On the final exam:** Question 2 (with Lesson 5), 15–22 marks. The biggest single block of the paper.

**Seen in:** Every year. 2023/24 Q2a, 2022/23 Q2a, 2021/22 Q2ab, 2018/19 Q2a, 2017/18, 2016/17 Q3a, 2015/16 Q2a…

**Study time:** 120 min (split around Zohor) | **Level for a beginner:** Medium — choosing the right variant is the skill

### The idea in plain words

**Now there are TWO separate groups** (brand X vs brand Y, men vs women, version A vs B) and we ask: is the difference between them real, or noise? Same 5 steps. The only new work is choosing among 4 formulas — and the question always tells you which one with a magic phrase.

**All the two-mean statistics have the same shape:** (difference of sample means − claimed difference) ÷ (standard error of the difference). The claimed difference Δ0 is almost always 0.

## The magic-phrase decoder

| Magic phrase in the question | Test | df |
|---|---|---|
| 'assume variances (σ) EQUAL' — small samples | Pooled t (compute $sp^2$ first) | n1 + n2 − 2 |
| 'variances NOT equal' — or nothing said, small samples | Welch T (big v formula) | v, ROUND DOWN |
| both samples LARGE (n ≥ 30) with s1, s2 given | z (use s1, s2 as σ1, σ2) | — |
| comparing two percentages / proportions | two-proportion z (pooled $\bar{p}$) | — |
| comparing consistency / variability / spread / sd's | $F = \frac{s_1^2}{s_2^2}$ | (n1−1, n2−1) |
| same people measured twice (before & after) | PAIRED t → Lesson 5 | n − 1 |

### TEMPLATE — pooled t (variances assumed equal)

$$H_0: \mu_1 = \mu_2 \,(i.e.\ \mu_1 - \mu_2 = 0)\ ,\ H_1: \mu_1 - \mu_2\ (\neq\ /\ <\ /\ >)\ \ 0$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \qquad t = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

3. df = n1 + n2 − 2 → t table.

4–5. Compare, decide, conclude in words.

### TEMPLATE — Welch T (variances not equal) — 2023/24 asked exactly this

$$T = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \rightarrow \quad ROUND\ \ DOWN$$

Everything else identical to any t test.

### TEMPLATE — two proportions z

$$\hat{p}_1 = \frac{x_1}{n_1}, \hat{p}_2 = \frac{x_2}{n_2}, \qquad \bar{p} = \frac{x_1 + x_2}{n_1 + n_2}, \qquad \bar{q} = 1 - \bar{p}$$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\bar{p}\bar{q}\ \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Critical from the z table. Everything else the same.

### TEMPLATE — F-test for two variances

$$H_0 : \sigma_1^2 = \sigma_2^2, H_1 : \sigma_1^2 \neq \sigma_2^2$$

(One-tailed version: H1 uses > — put the suspected-larger variance on top.)

$$F = \frac{s_1^2}{s_2^2} (SQUARE\ \ the\ \ sd's!)$$

3. df = (n1 − 1, n2 − 1) = (numerator, denominator). Two-tailed → use the F table for α/2 (e.g. α = 0.05 → the 0.025 table).

4–5. F > critical → reject. Conclude.

## WORKED EXAMPLE A — pooled t (real 2021/22 Q2a: batteries)

Brand X: n = 10, $\bar{x}$ = 325 min, s = 4. Brand Y: n = 10, $\bar{y}$ = 482, s = 6. Claim: X lasts SHORTER than Y. α = 0.05, variances assumed equal.

### MODEL ANSWER — batteries (pooled t, left-tailed)

**Step 1:** left-tailed:

$$H_0 : \mu_X = \mu_Y, H_1 : \mu_X < \mu_Y$$

**Step 2:** pooled variance first:

$$s_p^2 = \frac{9(16) + 9(36)}{18} = \frac{468}{18} = 26.000, s_p = 5.099$$

$$t = \frac{325 - 482}{5.099 \times \sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{-157}{2.280} = -68.849$$

**Step 3:** df = 18, α = 0.05 one-tail → critical −1.734.

**Step 4:** −68.849 < −1.734 → Reject H0 (massively).

**Step 5:** At the 0.05 level, there is sufficient evidence that brand X batteries last shorter than brand Y.

## WORKED EXAMPLE B — Welch T (real 2023/24 Q2a: product versions)

Version A: n = 50, $\bar{x}$ = 4.2, s = 0.8. Version B: n = 60, $\bar{x}$ = 4.5, s = 0.7. Is satisfaction DIFFERENT? α = 0.05. (Paper asked: hypotheses, statistic, df, decision.)

### MODEL ANSWER — versions (Welch, two-tailed)

**(i)** two-tailed:

$$H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$$

**(ii)** the two variance pieces:

$$\frac{s_1^2}{n_1} = \frac{0.64}{50} = 0.0128, \frac{s_2^2}{n_2} = \frac{0.49}{60} = 0.00817$$

$$T = \frac{4.2 - 4.5}{\sqrt{0.0128 + 0.00817}} = \frac{-0.3}{0.145} = -2.072$$

**(iii)** degrees of freedom:

$$v = \frac{(0.0128 + 0.00817)^2}{\frac{0.0128^2}{49} + \frac{0.00817^2}{59}} = \frac{0.000440}{0.00000447} = 98.3 \quad \rightarrow \quad df = 98$$

**(iv)** Critical ±t(0.025, 98) ≈ ±1.984. Since −2.072 < −1.984 → Reject H0. At the 0.05 level there IS a difference in mean satisfaction between the versions.

## WORKED EXAMPLE C — F-test (real 2022/23 Q2a: life satisfaction)

Older adults: n = 13, s = 1.21. Younger: n = 16, s = 0.63. Different VARIABILITY? α = 0.05.

### MODEL ANSWER — variability (F, two-tailed)

**Step 1:** two variances:

$$H_0: \sigma_1^2 = \sigma_2^2, H_1: \sigma_1^2 \neq \sigma_2^2$$

**Step 2:** larger variance on top:

$$F = \frac{1.21^2}{0.63^2} = \frac{1.464}{0.397} = 3.689$$

**Step 3:** df = (12, 15). Two-tailed α = 0.05 → F(0.025) table → critical ≈ 2.96.

**Step 4–5:** 3.689 > 2.96 → Reject H0. There IS a difference in variability between the two groups. (Tip: put the LARGER s on top as s1 so F > 1 and you only check the right tail.)

**TRAPS — where students lose marks**

Read the magic phrase FIRST. 'Assume variances equal' → pooled. 'Not equal' → Welch. Wrong choice = wrong df = wrong critical value.

Welch df v: ROUND DOWN (98.3 → 98), never up.

Two proportions: the pooled $\bar{p}$ uses the COUNTS (x1+x2)/(n1+n2), not the average of $\hat{p}1$ and $\hat{p}2$.

F-test: square the standard deviations. F table for two-tailed α = 0.05 is the 0.025 table.

Keep group 1 and group 2 in the SAME order everywhere (means, H1 direction, computation).

If both n ≥ 30, just use z with s1, s2 — no df needed (2015/16 Q2a style).

## PRACTICE · LEVEL 1 (warm-up, real 2015/16 Q2a — large samples z)

Age 18–34: n = 150, $\bar{x}$ = RM75.50, s = 53.20. Age 35+: n = 250, $\bar{x}$ = RM65.20, s = 46.10. Any difference in mean spending? α = 0.05 (variances not equal, but both n large → z).

**MODEL ANSWER — Level 1**

$$H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$$

$$z = \frac{75.5 - 65.2}{\sqrt{\frac{53.2^2}{150} + \frac{46.1^2}{250}}} = \frac{10.3}{\sqrt{18.868 + 8.501}} = \frac{10.3}{5.231} = 1.969$$

Critical ±1.960. Since 1.969 > 1.960 (just barely!) → Reject H0. There is a difference in mean spending between the age groups. (Notice how close — write 3 decimals or you get it wrong.)

## PRACTICE · LEVEL 2 (real exam: 2017/18 S1 Q3a + 2021/22 Q2b)

(a) Brand A: 12 cars, $\bar{x}$ = 27.2 mpg, s = 3.8. Brand B: 9 cars, $\bar{x}$ = 32.1, s = 4.3. At α = 0.01, does brand B have HIGHER mileage? Variances assumed equal.

(b) University A: 102 of 150 students favour a new grading system. University B: 108 of 180. Different proportions? α = 0.01.

**MODEL ANSWER — Level 2**

**(a)** H0: μA = μB, H1: μA < μB (B higher).

$$s_p^2 = \frac{11(3.8^2) + 8(4.3^2)}{19} = \frac{306.76}{19} = 16.145, s_p = 4.018$$

$$t = \frac{27.2 - 32.1}{4.018\sqrt{\frac{1}{12} + \frac{1}{9}}} = \frac{-4.9}{1.772} = -2.766$$

df = 19, critical −t(0.01,19) = −2.539. −2.766 < −2.539 → Reject H0 → brand B mileage is higher.

**(b)** $\hat{p}1$ = 102/150 = 0.680, $\hat{p}2$ = 108/180 = 0.600, $\bar{p}$ = 210/330 = 0.636, $\bar{q}$ = 0.364.

$$z = \frac{0.680 - 0.600}{\sqrt{0.636 \times 0.364 \times \left(\frac{1}{150} + \frac{1}{180}\right)}} = \frac{0.080}{0.053} = 1.504$$

Two-tailed α = 0.01 → ±2.576. 1.504 inside → Fail to reject H0. No significant difference in the

proportions.

## PRACTICE · LEVEL 3 (monster: 2014/15 Q2a — nonzero Δ0 + 2017/18 S2 Q3b F-test raw data)

(a) With tutorials: 11 students, $\bar{x}$ = 85, s = 4.7. Without: 17 students, $\bar{x}$ = 79, s = 6.1. Is there evidence the tutorial raises the average by MORE than 8 points? α = 0.05, variances equal.

(b) Line 1 alcohol content: 0.48, 0.39, 0.42, 0.52, 0.40. Line 2: 0.38, 0.37, 0.39, 0.41, 0.38. Test H0: σ1 = σ2 vs H1: σ1 ≠ σ2 at α = 0.05.

### MODEL ANSWER — Level 3

**(a)** Here Δ0 = 8 (not zero!):

$$H_0: \mu_1 - \mu_2 = 8 \ , H_1: \mu_1 - \mu_2 > 8$$

$$s_p^2 = \frac{10(4.7^2) + 16(6.1^2)}{26} = \frac{816.26}{26} = 31.395 \ , s_p = 5.603$$

$$t = \frac{(85 - 79) - 8}{5.603\sqrt{\frac{1}{11} + \frac{1}{17}}} = \frac{-2}{2.169} = -0.922$$

df = 26, critical t(0.05,26) = 1.706. −0.922 < 1.706 → Fail to reject H0. Not sufficient evidence that tutorials raise the average by more than 8 points.

**(b)** Calculator: $s1^2$ = 0.00312 (s1 = 0.056), $s2^2$ = 0.00023 (s2 = 0.015).

$$F = \frac{0.00312}{0.00023} = 13.565$$

df = (4,4); two-tailed α = 0.05 → F(0.025)(4,4) = 9.60. 13.565 > 9.60 → Reject H0: line 1 is NOT as consistent as line 2.

**Monster factor:** (a) subtracting Δ0 = 8 inside the numerator — the single rarest twist they use; (b) raw data F-test where you build both variances yourself.

# LESSON 5 PAIRED t-TEST (BEFORE vs AFTER)

**On the final exam:** Inside Question 2, 8–10 marks.

**Seen in:** Appears in EVERY single past paper since 2014. A guaranteed question.

**Study time:** 50 min | **Level for a beginner:** Easy — one column of subtractions then a 1-sample t

## The idea in plain words

**Same people (or objects) measured twice** — before and after a program, drug, training. The groups are NOT independent, so Lesson 4 is illegal here. The fix is beautiful: subtract each pair to get ONE column of differences D, then run an ordinary one-sample t-test on that column with μD = 0.

**How to spot it:** the data table has matching columns for the SAME individuals ('before/after', 'round 3/round 4', '1 hour/24 hours'). If each row is one person measured twice → paired.

## TEMPLATE — paired t

1. Define D = (before − after) or (after − before) — CHOOSE the direction that makes the claim positive, and say which you chose.

$$H_0: \mu_D = 0 \ , H_1: \mu_D > 0 \quad (\ also \ < \ , \neq \ )$$

$$\bar{D} = \frac{\Sigma D}{n}, s_D = sd \ of \ the \ D \ column \ , \qquad t = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}$$

3. df = n − 1 (n = number of PAIRS).

4–5. Compare with t table, decide, conclude.

## WORKED EXAMPLE — the real 2023/24 Q2(b): tutoring program

Scores before: 72, 68, 75, 80, 78. After: 82, 85, 88, 90, 86. Does tutoring significantly IMPROVE scores? α = 0.05.

### MODEL ANSWER — tutoring (paired t, right-tailed)

**Step 1:** Let D = after − before (improvement positive).

$$H_0: \mu_D = 0 \ , H_1: \mu_D > 0$$

**Step 2:** D = 10, 17, 13, 10, 8. $\bar{D}$ = 58/5 = 11.600. sD = 3.507.

$$t = \frac{11.600}{3.507 / \sqrt{5}} = \frac{11.600}{1.568} = 7.396$$

**Step 3:** df = 4, α = 0.05 one-tail → critical 2.132.

**Step 4:** 7.396 > 2.132 → Reject H0.

**Step 5:** At the 0.05 level, the tutoring program has a significant positive effect on test scores.

**TRAPS — where students lose marks**

n in the formula = number of PAIRS (5 above), not 10.

State your D direction clearly, then keep it. If you use D = before − after for an 'improvement' claim, H1 becomes μD < 0 — also fine, just be consistent.

sD is the sd of the differences column ONLY. Never touch the original columns' sd's.

If the exam GIVES you the hypotheses (2017/18 gave H1: μD < 0 with D = before − after), follow THEIR direction.

A tiny $\bar{D}$ can still 'fail to reject' — that is a valid answer (see iron example below).

## PRACTICE · LEVEL 1 (warm-up, real 2014/15 Q2b: golf)

Round 3: 76, 75, 72, 75, 79. Round 4: 70, 73, 71, 68, 76. Any difference in mean score? α = 0.05. (First find $\bar{D}$ and sD, then test.)

**MODEL ANSWER — Level 1**

D = round3 − round4 = 6, 2, 1, 7, 3. $\bar{D}$ = 19/5 = 3.800, sD = 2.588.

$$t = \frac{3.800}{2.588/\sqrt{5}} = \frac{3.800}{1.157} = 3.283$$

Two-tailed, df = 4 → ±2.776. Since 3.283 > 2.776 → Reject H0: the mean scores of the two rounds differ.

## PRACTICE · LEVEL 2 (real exam: 2021/22 Q2c = 2016/17 Q3b, and 2022/23 Q2b)

(a) Weight before: 95, 81, 76, 96, 82, 87, 71, 82, 77, 83, 81, 65. After: 89, 77, 77, 94, 80, 86, 72, 82, 74, 81, 78, 63. Is the exercise program effective (reduces weight)? α = 0.01.

(b) Iron content before: 11.6, 18.7, 15.9, 20.6, 10.1, 17.4, 7.2, 12.2, 11.7. After: 10.0, 21.6, 13.9, 22.8, 11.5, 16.2, 8.1, 10.8, 9.6. Any difference? α = 0.05.

**MODEL ANSWER — Level 2**

**(a)** D = before − after = 6, 4, −1, 2, 2, 1, −1, 0, 3, 2, 3, 2. $\bar{D}$ = 23/12 = 1.917, sD = 2.021.

$$t = \frac{1.917}{2.021/\sqrt{12}} = \frac{1.917}{0.583} = 3.286$$

H1: μD > 0, df = 11, critical t(0.01,11) = 2.718. 3.286 > 2.718 → Reject H0 → the program is effective. (This exact question appeared in TWO different years.)

**(b)** D = 1.6, −2.9, 2.0, −2.2, −1.4, 1.2, −0.9, 1.4, 2.1. $\bar{D}$ = 0.9/9 = 0.100, sD = 1.946.

$$t = \frac{0.100}{1.946/\sqrt{9}} = \frac{0.100}{0.649} = 0.154$$

Two-tailed, df = 8 → ±2.306. 0.154 inside → Fail to reject H0: no significant difference in iron content.

## PRACTICE · LEVEL 3 (monster: 2015/16 Q2b — the Δ0 = 3 twist)

Words recalled 1 hour later: 14, 12, 18, 7, 11, 9, 16, 15, 13, 17. After 24 hours: 10, 4, 14, 5, 8, 5, 11, 12, 9, 10. Is the 1-hour mean recall MORE THAN 3 words above the 24-hour recall? α = 0.05.

**MODEL ANSWER — Level 3**

D = 1hr − 24hr = 4, 8, 4, 2, 3, 4, 5, 3, 4, 7. $\bar{D}$ = 44/10 = 4.400, sD = 1.838.

**The twist:** 'by more than 3' moves the claimed difference to 3 — subtract it in the numerator:

$$H_0: \mu_D = 3 \quad , H_1: \mu_D > 3$$

$$t = \frac{4.400 - 3}{1.838/\sqrt{10}} = \frac{1.400}{0.581} = 2.409$$

df = 9, critical t(0.05,9) = 1.833. 2.409 > 1.833 → Reject H0. Yes — mean recall after 1 hour exceeds the 24-hour recall by more than 3 words.

**Monster factor:** spotting that 'by more than 3' shifts μD from 0 to 3. If you can do this, no paired question can touch you.

# LESSON 6 CHI-SQUARE: GOODNESS-OF-FIT & INDEPENDENCE

**On the final exam:** Question 3, 20–25 marks. The single biggest question on most papers.

**Seen in:** Every year: 2013/14 Q4, 2014/15 Q3, 2015/16 Q3, 2016/17 Q4, 2017/18 Q4, 2018/19 Q3, 2021/22 Q3, 2022/23 Q3, 2023/24 Q3

**Study time:** 90 min (split around Asar) | **Level for a beginner:** Medium — bookkeeping, not brains. Tables win marks.

## The idea in plain words

**Now the data are COUNTS in categories** (customers per day, donors per blood group). One idea powers everything: compare the Observed counts (o) with the Expected counts (e) that the claim predicts. If o and e are close → claim survives. Far apart → reject. Distance is measured by:

$$\chi^2 = \Sigma \frac{(o - e)^2}{e}$$

**This test is ALWAYS right-tailed — one critical value; reject when $\chi^2$ is big.**

**Type 1 — Goodness-of-fit (GOF):** ONE row of categories. 'Are reservations equal across 5 days?' → e = total/k each. 'Blood groups follow 44%, 45%, 8%, 3%?' → e = total × each %. df = k − 1.

**Type 2 — Independence:** a TABLE of two variables (age × internet usage). 'Is A related to B?' H0: independent (not related); H1: related. e for each cell = (row total × column total)/grand total. df = (r − 1)(c − 1).

## TEMPLATE — chi-square GOF

1. H0: the distribution is as claimed (all equal, or the given %). H1: at least one differs.
2. e = total/k (equal case) or e = total × percentage (given-% case). Make a small o/e table.

$$\chi^2 = \Sigma \frac{(o - e)^2}{e} \text{ — show the first two terms at minimum}$$

3. df = k − 1, right tail, column α → critical value.
4–5. $\chi^2$ > critical → Reject. Conclude about the claim in words.

## TEMPLATE — chi-square independence

1. H0: variable A and variable B are independent (NOT related). H1: they are related.
2. Row totals, column totals, grand total. Then EVERY cell: e = (row total × column total)/grand total. Write the e-table.

$$\chi^2 = \Sigma \frac{(o - e)^2}{e} \text{ across all cells}, \qquad e = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

3. df = (r − 1)(c − 1), column α → critical.
4–5. Decide + words: 'there is (no) evidence of a relationship between A and B.'

## WORKED EXAMPLE A — GOF (real 2023/24 Q3a: reservations)

Reservations Mon–Fri: 30, 25, 35, 28, 40. Are reservations uniform across weekdays? α = 0.05.

### MODEL ANSWER — reservations (GOF)

**Step 1:** H0: reservations are equally distributed across the 5 days. H1: at least one day differs.

**Step 2:** total = 158 → e = 158/5 = 31.6 for every day.

$$\chi^2 = \frac{(30-31.6)^2}{31.6} + \frac{(25-31.6)^2}{31.6} + \frac{(35-31.6)^2}{31.6} + \frac{(28-31.6)^2}{31.6} + \frac{(40-31.6)^2}{31.6}$$

$$= \frac{2.56 + 43.56 + 11.56 + 12.96 + 70.56}{31.6} = \frac{141.2}{31.6} = 4.468$$

**Step 3:** df = 5 − 1 = 4, α = 0.05 → critical 9.488.

**Step 4:** 4.468 < 9.488 → Fail to reject H0.

**Step 5:** At the 0.05 level, the data do NOT disprove that reservations are equally distributed across weekdays. (Yes — the newest paper's answer was 'fail to reject'.)

## WORKED EXAMPLE B — Independence (real 2023/24 Q3b: age × internet)

159 adults. Rows: Never (38, 31), Sometimes (31, 4), Every day (50, 5) — columns are Age 25–64 and 65+. Related at α = 0.01?

### MODEL ANSWER — internet (independence)

**Step 1:** H0: age and internet usage are independent. H1: they are related.

**Step 2:** Row totals 69, 35, 55; column totals 119, 40; grand 159. Expected counts:

| cell | Age 25–64 | Age 65+ |
|---|---|---|
| Never (69) | 69×119/159 = 51.642 | 69×40/159 = 17.358 |
| Sometimes (35) | 35×119/159 = 26.195 | 35×40/159 = 8.805 |
| Every day (55) | 55×119/159 = 41.164 | 55×40/159 = 13.836 |

$$\chi^2 = \frac{(38-51.642)^2}{51.642} + \frac{(31-17.358)^2}{17.358} + \frac{(31-26.195)^2}{26.195}$$

$$+ \frac{(4-8.805)^2}{8.805} + \frac{(50-41.164)^2}{41.164} + \frac{(5-13.836)^2}{13.836}$$

$$= 3.604 + 10.722 + 0.881 + 2.622 + 1.897 + 5.642 = 25.368$$

**Step 3:** df = (3−1)(2−1) = 2, α = 0.01 → critical 9.210.

**Step 4–5:** 25.368 > 9.210 → Reject H0. At the 1% level, a relationship exists between age and internet usage.

### TRAPS — where students lose marks

Expected counts keep decimals (31.6, 51.642). NEVER round e to whole numbers.

Check: Σe must equal Σo (158 = 158). If not, you made an arithmetic slip.

df for GOF uses k = number of CATEGORIES (not the total count!). df for tables: (r−1)(c−1).

$\chi^2$ tests here are ALWAYS right-tailed. One critical value only.

'Test the claim that they are NOT related' — careful: H0 is STILL 'independent'. Rejecting H0 = they ARE related.

Sometimes the raw data comes as a list (2022/23 jogging) — first build the r×c table, marks are given for it.

**'Test the HOMOGENEITY among the shops' (2017/18):** identical computation to independence — same e = row×col/total, same df = (r−1)(c−1). Only the wording changes: H0: the distribution is the SAME in every group; H1: at least one group differs. (For the 2017/18 shops table: $\chi^2 = 15.168 > 9.488$ → reject → the shops' repair distributions are not homogeneous.)

## PRACTICE · LEVEL 1 (warm-up, real 2018/19 Q3a: donuts)

Observed preference: plain 13, strawberry 12, blueberry 16, cinnamon 19. Are all types equally preferred? α = 0.05. (State hypotheses, critical value, statistic, conclusion — the exam's exact 4 sub-parts.)

### MODEL ANSWER — Level 1

H0: preferences equal; H1: at least one differs. Total 60 → e = 15 each.

$$\chi^2 = \frac{4 + 9 + 1 + 16}{15} = \frac{30}{15} = 2.000$$

df = 3 → critical 7.815. 2.000 < 7.815 → Fail to reject H0. Customer preference is similar across all four types.

## PRACTICE · LEVEL 2 (real exam: 2022/23 Q3a + 2016/17 Q4a)

(a) Accidents Mon–Sun: 28, 21, 15, 14, 32, 23, 19. Equally distributed over the week? 95% confidence.

(b) Blood donors: O 67, A 83, B 29, AB 8 (n = 187). Expected: O 44%, A 45%, B 8%, AB 3%. Different from expected? 95% confidence.

### MODEL ANSWER — Level 2

**(a)** total 152 → e = 152/7 = 21.714 each. $\chi^2 = 11.947$. df = 6 → critical 12.592. Since 11.947 < 12.592 → Fail to reject H0 — no evidence the accidents are unequally distributed. (A famous trap: it LOOKS unequal, but the test says the evidence is not enough — by 0.6!)

**(b)** e = 187×0.44 = 82.28; 187×0.45 = 84.15; 187×0.08 = 14.96; 187×0.03 = 5.61.

$$\chi^2 = \frac{(67-82.28)^2}{82.28} + \frac{(83-84.15)^2}{84.15} + \frac{(29-14.96)^2}{14.96} + \frac{(8-5.61)^2}{5.61}$$

$$= 2.838 + 0.016 + 13.176 + 1.018 = 17.048$$

df = 3 → critical 7.815. 17.048 > 7.815 → Reject H0: the donor distribution differs from the expected distribution.

## PRACTICE · LEVEL 3 (monster: 2021/22 Q3 — the full 20-mark question)

(a) Conceptual: parcels across 5 departments, H0: p1 = … = p5. (i) $\chi^2 = 16$ at α = 0.01 — conclusion? (ii) $\chi^2 = 18$ at α = 0.05? (iii) only 4 departments, $\chi^2 = 19$ at α = 0.025?

(b) Gender gap: Men (325 support, 151 oppose, 324 undecided), Women (258, 241, 251). α = 0.05. Related?

### MODEL ANSWER — Level 3

**(a-i)** df = 4, $\chi^2(0.01) = 13.277$. 16 > 13.277 → Reject H0.

**(a-ii)** df = 4, $\chi^2(0.05) = 9.488$. 18 > 9.488 → Reject H0.

**(a-iii)** df = 3, $\chi^2(0.025) = 9.348$. 19 > 9.348 → Reject H0. (The whole sub-question is just table reading!)

**(b)** Row totals 800, 750; column totals 583, 392, 575; n = 1550. Expected: men 300.903, 202.323, 296.774; women 282.097, 189.677, 278.226.

$$\chi^2 = 1.930 + 13.020 + 2.497 + 2.058 + 13.887 + 2.663 = 36.056$$

df = (2−1)(3−1) = 2 → critical 5.991. 36.056 > 5.991 → Reject H0: gender IS related to opinion — the president faces a gender gap.

**Recycled alert:** these exact numbers appeared in BOTH 2013/14 and 2021/22.

# LESSON 7 CORRELATION (PEARSON r & SPEARMAN rs)

**On the final exam:** Question 4, 15–25 marks. Long but 100% mechanical.

**Seen in:** Every year: 2023/24 Q4, 2022/23 Q4, 2021/22 Q4, 2018/19 Q4a, 2017/18, 2016/17 Q5, 2015/16 Q4a, 2014/15 Q4a

**Study time:** 75 min | **Level for a beginner:** Easy-medium — the formula looks scary but is pure bookkeeping

## The idea in plain words

**Correlation r answers:** how strongly do two things move together in a straight line? r is always between −1 and +1. Sign = direction (+ move together, − opposite). Size = strength: |r| near 1 strong, near 0.5 moderate, near 0 weak/none.

**The whole computation is 5 sums.** From the data build Σx, Σy, Σxy, Σx², Σy² — then feed the formula. Most marks are for the sums table, so write it.

**Pearson vs Spearman:** Pearson for normal numeric data. Spearman when the data are RANKS (1st, 2nd, 3rd…) or clearly not normal / has a big outlier. Exams love asking 'which is suitable? justify.'

**Scatter plot sketch:** just plot the pairs roughly on labelled axes and say 'points rise together → positive relationship' (or fall → negative). 2 easy marks.

## TEMPLATE — Pearson r

1. Table with columns x, y, xy, x², y² → totals give Σx, Σy, Σxy, Σx², Σy² (some may be GIVEN — use them!).

$$r = \frac{\Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}}{\sqrt{\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)\left(\Sigma y^2 - \frac{(\Sigma y)^2}{n}\right)}}$$

3. Interpret: sign + strength + context. 'r = 0.720 → fairly strong positive linear relationship between blood pressure and albumin.'

## TEMPLATE — Spearman rs (ranked data)

1. Rank the x values (1 = largest or 1 = smallest — same rule for both variables). Rank the y values.

$$r_s = 1 - \frac{6\Sigma d^2}{n(n^2 - 1)}, \qquad d = rank(x) - rank(y)$$

3. Interpret exactly like r.

## TEMPLATE — Is r significant? (the t-test for correlation)

$$H_0:\ \rho = 0 \quad (no\ \ correlation)\ ,H_1:\ \rho \neq 0$$

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}}, \qquad df = n - 2$$

3–5. Compare with ±t(α/2, n−2); reject → 'the correlation is significant.'

## WORKED EXAMPLE — the real 2023/24 Q4: blood pressure & albumin

7 patients. p: 160, 162, 175, 185, 178, 170, 182. a: 22, 29, 31, 35, 33, 38, 45. Given $\Sigma p^2 = 210402$, $\Sigma a^2 = 8069$. (a) scatter + direction, (b) compute r and interpret, (c) if medication LOWERS p, what happens to a?

### MODEL ANSWER — blood pressure & albumin

**(a)** Sketch: x-axis p, y-axis a; plot the 7 points; they drift upward → a positive relationship is observed.

**(b)** Build the missing sums: $\Sigma p = 1212$, $\Sigma a = 233$, $\Sigma pa = 160\times22 + 162\times29 + \ldots + 182\times45 = 40642$. n = 7.

$$numerator = 40642 - \frac{(1212)(233)}{7} = 40642 - 40342.286 = 299.714$$

$$denominator = \sqrt{\left(210402 - \frac{1212^2}{7}\right)\left(8069 - \frac{233^2}{7}\right)} = \sqrt{524.286 \times 313.429} = 416.271$$

$$r = \frac{299.714}{416.271} = 0.720$$

Interpretation: fairly strong positive linear relationship — higher blood pressure goes with higher albumin.

**(c)** Since r is positive, lowering p is expected to LOWER a as well (they move in the same direction).

### TRAPS — where students lose marks

If the paper GIVES you $\Sigma x^2$ or $\Sigma y^2$, do not recompute them — that is a gift (and a time bomb if you re-add wrongly).

$(\Sigma x)^2$ means 'total, then square'. $\Sigma x^2$ means 'square each, then total'. Mixing these is the #1 correlation error.

Keep 3 decimals inside the parts (numerator/denominator) and for the final r.

r measures STRAIGHT-LINE relation only. r ≈ 0 does not mean 'no relationship at all', it means no LINEAR one.

Correlation is NOT causation — if asked to comment, say a third factor could drive both.

Spearman with tied values: give each tied value the AVERAGE of the ranks they occupy.

## PRACTICE · LEVEL 1 (warm-up, hand-sized)

x: 1, 2, 3, 4, 5. y: 5, 3, 4, 2, 1. Compute r and interpret.

**MODEL ANSWER — Level 1**

Sums table: Σx = 15, Σy = 15, Σxy = (1×5) + (2×3) + (3×4) + (4×2) + (5×1) = 5 + 6 + 12 + 8 + 5 = 36, $\Sigma x^2 = 55$, $\Sigma y^2 = 55$.

$$numerator = 36 - \frac{(15)(15)}{5} = -9, \qquad denominator = \sqrt{(55 - 45)(55 - 45)} = \sqrt{100}$$

$$r = \frac{-9}{10} = -0.900$$

Strong negative linear relationship: as x rises, y falls. (Habit to copy: after building the sums, re-add each column once — a single wrong product poisons everything downstream.)

## PRACTICE · LEVEL 2 (real exam: 2022/23 Q4 + 2021/22 Q4)

(a) Rainfall x: 4.3, 4.5, 5.9, 5.6, 6.1, 5.2, 3.8, 2.1, 7.5. Particulate removed y: 126, 121, 116, 118, 114, 118, 132, 141, 108. Given $\Sigma x^2 = 244.26$, $\Sigma y^2 = 133786$. Sketch, compute r, check consistency with the sketch.

(b) Stress x: 12, 19, 10, 14, 8, 5, 15, 7, 11. Satisfaction y: 4, 1, 5, 6, 8, 9, 4, 8, 6. Which coefficient type suits this data (justify), then compute r and conclude.

**MODEL ANSWER — Level 2**

**(a)** Σx = 45, Σy = 1094, Σxy = 5348.2, n = 9.

$$numerator = 5348.2 - \frac{45 \times 1094}{9} = 5348.2 - 5470 = -121.8$$

$$denominator = \sqrt{(244.26 - 225)(133786 - 132987.111)} = \sqrt{19.26 \times 798.889} = 124.456$$

Interpretation: very strong NEGATIVE linear relationship — more rainfall, less particulate remaining. Yes, consistent with the downward scatter.

**(b)** Both variables are numeric scores → Pearson is suitable (data assumed approximately normal). Σx = 101, Σy = 51, Σxy = 492, $\Sigma x^2 = 1285$, $\Sigma y^2 = 339$.

$$numerator = 492 - \frac{101 \times 51}{9} = -80.333, \qquad denominator = \sqrt{151.556 \times 50} = 87.0$$

$$r = \frac{-80.333}{87.050} = -0.923$$

Strong negative — more stress, less satisfaction.

## PRACTICE · LEVEL 3 (monster: Spearman 2015/16 Q4a + significance test 2016/17 Q5)

(a) Transport manager ranks 8 car models: S=5, T=1, U=7, V=2, W=8, X=6, Y=4, Z=3. Prices: 49000, 70000, 56000, 73000, 52000, 57300, 68300, 65000. Compute Spearman rs and comment.

(b) For n = 8 pairs with r = 0.842 (plastic usage vs household size): is the correlation significant at 95% confidence? Would the decision change at 99%?

**MODEL ANSWER — Level 3**

**(a)** Rank prices with 1 = most expensive: V(73000)=1, T(70000)=2, Y(68300)=3, Z(65000)=4, X(57300)=5, U(56000)=6, W(52000)=7, S(49000)=8.

d = manager − price rank: S: 5−8 = −3; T: 1−2 = −1; U: 7−6 = 1; V: 2−1 = 1; W: 8−7 = 1; X: 6−5 = 1;

Y: 4−3 = 1; Z: 3−4 = −1. $\Sigma d^2 = 9+1+1+1+1+1+1+1 = 16$.

$$r_s = 1 - \frac{6(16)}{8(64-1)} = 1 - \frac{96}{504} = 0.810$$

Strong positive agreement: the manager's ranking largely follows price (rank 1 ≈ most expensive).

**(b)** test whether the correlation is significant:

$$H_0: \rho = 0 \;, H_1: \rho \neq 0$$

$$t = \frac{0.842}{\sqrt{\frac{1-0.842^2}{6}}} = 3.829 \;, \qquad df = 6$$

At 95%: ±2.447 → reject → significant. At 99%: ±3.707 → 3.829 still > 3.707 → still significant (but only just).

**Monster factor:** (a) building ranks yourself and keeping the direction consistent; (b) re-testing at a stricter α — check the NEW critical value before answering.

# LESSON 8 LINEAR REGRESSION (LEAST SQUARES LINE)

**On the final exam:** Question 5, 10–20 marks.

**Seen in:** Every year: 2023/24 Q5, 2022/23 Q5 (R output!), 2021/22 Q5, 2018/19 Q4b, 2017/18 Q5, 2016/17 Q6, 2015/16 Q4b, 2014/15 Q4b

**Study time:** 65 min | **Level for a beginner:** Easy-medium — reuses the exact sums from Lesson 7

## The idea in plain words

**Regression draws the best straight line through the points:** ŷ = b0 + b1x. b1 (slope, الميل) = how much y changes when x rises by 1. b0 (intercept) = predicted y when x = 0. Once you have the line, prediction is just substitution.

**You already did the hard part in Lesson 7:** the same five sums Σx, Σy, Σxy, Σx², Σy² feed b1. Exam Q4 and Q5 often share sums — compute once, use twice.

**R² (coefficient of determination):** the % of variation in y explained by x. $R^2 = r^2$. If r = 0.995, $R^2$ = 0.990 → '99% of the variation in service time is explained by the number of copiers'.

**Slope significance test:** 'does x really affect y?' → t = b1/sb1 with df = n − 2 (sb1 is usually GIVEN). Reject → x significantly affects y.

## TEMPLATE — least squares regression

$$b_1 = \frac{\Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

1. Sums table (or reuse Lesson 7's). 2. b1 (3 decimals) → b0. 3. Write the line: ŷ = b0 + b1x.

4. Predict: substitute the given x into the line.

5. If asked about a far-away x (outside the data range): say prediction is NOT appropriate — extrapolation is unreliable.

6. Slope test (does x really affect y?):

$$H_0: \beta_1 = 0, \quad H_1: \beta_1 \neq 0, \qquad t = \frac{b_1}{s_{b1}}, \qquad df = n - 2; \qquad reject \rightarrow x \ affects \ y$$

## WORKED EXAMPLE — the recycled classic (2016/17 Q6 = 2021/22 Q5a): engine volume & mileage

9 vehicles. x (engine volume): 97, 85, 98, 105, 120, 151, 140, 134, 146. y (mileage): 24, 29, 26, 24, 24, 22, 23, 23, 21. (i) the five sums, (ii) the least-squares equation, (iii) predict mileage at x = 125 (and at 160).

### MODEL ANSWER — engine & mileage

**(i)** Σx = 1076, Σy = 216, Σxy = 25431, $\Sigma x^2$ = 133336, $\Sigma y^2$ = 5228, n = 9.

**(ii)**

$$b_1 = \frac{25431 - \frac{1076 \times 216}{9}}{133336 - \frac{1076^2}{9}} = \frac{25431 - 25824}{133336 - 128641.778} = \frac{-393}{4694.222} = -0.084$$

$$b_0 = \bar{y} - b_1\bar{x} = 24 - (-0.084)(119.556) = 34.009$$

(where $\bar{x}$ = 1076/9 = 119.556 and $\bar{y}$ = 216/9 = 24)

$$\hat{y} = 34.009 - 0.084x$$

**(iii)** ŷ(125) = 34.009 − 0.084(125) = 34.009 − 10.465* = 23.544 km. ŷ(160) = 20.614 km. (*keep b1's full digits −0.08372 in the calculator; 3-decimal display only at the end.)

## Reading R output (2022/23 Q5 — they show a printout, you read numbers)

The R printout labels: 'Intercept' Estimate = b0; the x-row Estimate = b1; 'Multiple R-squared' = $R^2$. For the Accu-Copiers data (11 calls): b0 = 11.464, b1 = 24.602, $R^2$ = 0.990.

### MODEL ANSWER — copiers (R output skills)

(a) b0 = 11.464, b1 = 24.602 (straight from the output).

(b) Interpretation: each extra copier adds ≈ 24.602 minutes of service time (b1). b0 = 11.464 min is the 'time for zero copiers' — a base/setup time; practically it has little meaning on its own.

(c) ŷ = 11.464 + 24.602x.

(d) ŷ(4) = 11.464 + 24.602(4) = 109.873 minutes.

(e) $R^2$ = 0.990 → 99.0% of the variation in minutes is explained by the number of copiers.

(f) Slope test:

$$H_0: \beta_1 = 0 \;, H_1: \beta_1 \neq 0$$

$$t = \frac{b_1}{s_{b1}} = \frac{24.602}{0.805} = 30.580$$

df = 9 → ±2.262. Reject H0 → strong evidence of a linear relationship.

### TRAPS — where students lose marks

b1 FIRST, then b0 (b0 needs b1). Keep b1's full calculator value when computing b0 and predictions.

Slope interpretation must include units and direction: '−0.084 → each +1 cubic unit of engine volume LOWERS mileage by 0.084 km'.

Prediction outside the data range (x = 15 g when data go 1.0–4.5) → refuse: extrapolation is not appropriate.

$R^2$ as a percent: multiply by 100 and SAY 'of the variation in y explained by x'.

Slope test df = n − 2 (not n − 1).

If the question gives sb1 (e.g. 0.07), use it directly — do not compute your own.

## PRACTICE · LEVEL 1 (warm-up, real 2017/18 S1 Q5)

Years worked x: 5, 3, 4, 10, 15. Hourly pay y: 25, 20, 21, 35, 38. Find r, b1, b0, the equation, and predict pay at x = 20.

### MODEL ANSWER — Level 1

Σx = 37, Σy = 139, Σxy = 1189, Σx² = 375, Σy² = 4135, n = 5.

$$b_1 = \frac{1189 - \frac{37 \times 139}{5}}{375 - \frac{37^2}{5}} = \frac{160.4}{101.2} = 1.585$$

$$b_0 = 27.8 - 1.585(7.4) = 16.071, \qquad \hat{y} = 16.071 + 1.585x$$

r = 160.4/√(101.2 × 270.8) = 0.969 (very strong positive). Prediction: ŷ(20) = 16.071 + 1.585(20) = 47.771.

(Bonus thought: x = 20 is slightly beyond the data range 3–15, so treat the prediction with caution.)

## PRACTICE · LEVEL 2 (real exam: 2023/24 Q5a + 2015/16 Q4b)

(a) 7 motorbikes. x: 97, 88, 105, 151, 140, 134, 118. y: 24, 21, 24, 25, 22, 23, 22. Find the five sums, the least-squares equation, and estimate mileage at x = 120.

(b) Fertilizer x: 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5. Yield y: 3.9, 4.4, 5.8, 6.6, 7.0, 7.1, 7.3, 7.7. Equation? Yield at 2.8 g? Is predicting at 15 g appropriate?

### MODEL ANSWER — Level 2

**(a)** Σx = 833, Σy = 161, Σxy = 19229, Σx² = 102459, Σy² = 3715.

$$b_1 = \frac{19229 - \frac{833 \times 161}{7}}{102459 - \frac{833^2}{7}} = \frac{70}{3332} = 0.021, \quad b_0 = 23 - 0.021(119) = 20.500$$

$$\hat{y} = 20.500 + 0.021x, \qquad \hat{y}(120) = 23.021 \ km$$

**(b)** Σx = 22, Σy = 49.8, Σxy = 148.3, Σx² = 71, n = 8.

$$b_1 = \frac{148.3 - \frac{22 \times 49.8}{8}}{71 - \frac{22^2}{8}} = \frac{11.35}{10.5} = 1.081, \quad b_0 = 6.225 - 1.081(2.75) = 3.252$$

$$\hat{y} = 3.252 + 1.081x, \qquad \hat{y}(2.8) = 6.279 \ kg$$

15 g: NOT appropriate — far outside the data range (1.0–4.5 g); the straight-line pattern may not continue (extrapolation).

## PRACTICE · LEVEL 3 (monster: 2023/24 Q5b = 2021/22 Q5b — the slope hypothesis test)

6 patients. Age t: 35, 74, 48, 60, 42, 26. Blood pressure p: 88, 130, 120, 135, 98, 80. (i) scatter + expected correlation; (ii) given sb1 = 0.07 and the fitted line, does age affect blood pressure? Test at 95% confidence.

**MODEL ANSWER — Level 3**

**(i)** Points rise strongly together → strong positive correlation expected (r = 0.919 if you compute it).

**(ii)** First the slope: Σt = 285, Σp = 651, Σtp = 32756, Σt² = 15045.

$$b_1 = \frac{32756 - \frac{285 \times 651}{6}}{15045 - \frac{285^2}{6}} = \frac{1833.5}{1507.5} = 1.216 \qquad (line: \ \hat{y} = 50.728 + 1.216t)$$

$$H_0: \beta_1 = 0 \quad (age \ does \ not \ affect \ BP), H_1: \beta_1 \neq 0$$

$$t = \frac{b_1}{s_{b1}} = \frac{1.216}{0.07} = 17.371, \qquad df = 6 - 2 = 4 \rightarrow critical \pm 2.776$$

17.371 > 2.776 → Reject H0. At 95% confidence, the patient's age DOES significantly affect blood pressure.

**Recycled alert:** the same 6 patients with sb1 = 0.07 appeared in 2021/22 AND 2023/24. Know this one cold.

# LESSON 9 ANOVA — COMPARING 3+ GROUP MEANS

**On the final exam:** Question 6, 10–20 marks. Often the last question — bank it and finish strong.

**Seen in:** Every year: 2023/24 Q6, 2022/23 Q6, 2021/22 Q6, 2018/19 Q5, 2017/18 Q6, 2016/17 Q7, 2015/16 Q5

**Study time:** 60 min | **Level for a beginner:** Looks scary, is actually an 8-step recipe with zero decisions

## The idea in plain words

**Three or more groups (drug A / B / C), one question:** are all the population means equal? Comparing pairs one-by-one is illegal (errors pile up). ANOVA compares two kinds of spread instead:

- **Variance BETWEEN samples** = how far apart the group means are ($n \cdot s^2_{\bar{x}}$ where $s^2_{\bar{x}}$ is the variance OF the means).
- **Variance WITHIN samples** = the average of the group variances ($s^2_p$) — the natural noise.

**F = between/within.** If the means truly are equal, F ≈ 1. A big F says the means differ more than noise can explain → reject. Your syllabus covers EQUAL group sizes only — every group has the same n.

**H1 is always: 'at least one mean is different'** — never 'all means are different'.

## TEMPLATE — one-way ANOVA — the 8 steps your slides use

$$H_0: \mu_1 = \mu_2 = \mu_3 (= \mu_4) \ , H_1: \ at \ least \ one \ mean \ is \ different$$

2. For each group: n, $\bar{x}$, s (calculator STAT mode; skip if the table already gives them).

3a. Grand mean $\bar{\bar{x}}$ = average of the group means. 3b. $s_{\bar{x}}$ = sd of the group means. 3c. Between = $n \cdot s^2_{\bar{x}}$.

4. Within: $s^2_p$ = average of the group variances = $(s1^2 + s2^2 + \ldots)/k$.

$$F = \frac{n s^2_{\bar{x}}}{s^2_p}$$

6. df: numerator = k − 1, denominator = k(n − 1).

7. Critical value from the F table for the given α.

8. F > critical → Reject H0 → at least one mean differs. Conclude in the story's words.

## WORKED EXAMPLE — the real 2022/23 Q6: mice & door colours

Times (min): Red 9, 11, 10, 9, 15 • Green 20, 21, 23, 17, 30 • Black 6, 5, 8, 14, 7. (a) mean and sd per colour, (b) F, (c) test at α = 0.01: are mice sensitive to colour?

### MODEL ANSWER — mice ANOVA

**Step 1:** compare all three means:

$$H_0: \mu_R = \mu_G = \mu_B, H_1: \text{ at least one mean is different}$$

**Step 2 (a):** Red: $\bar{x}$ = 10.8, s = 2.490 ($s^2$ = 6.2). Green: $\bar{x}$ = 22.2, s = 4.868 ($s^2$ = 23.7). Black: $\bar{x}$ = 8.0, s = 3.536 ($s^2$ = 12.5). k = 3, n = 5.

**Step 3:** $s_{\bar{x}}$ = sd of the three means {10.8, 22.2, 8.0} = 7.522 → $s^2_{\bar{x}}$ = 56.573.

$$\bar{\bar{x}} = \frac{10.8 + 22.2 + 8.0}{3} = 13.667, \quad \text{between} = n s^2_{\bar{x}} = 5 \times 56.573 = 282.867$$

**Step 4:** within:

$$s_p^2 = \frac{6.2 + 23.7 + 12.5}{3} = 14.133$$

**Step 5:** test statistic:

$$F = \frac{282.867}{14.133} = 20.014$$

**Step 6–7:** df = (k−1, k(n−1)) = (2, 12); α = 0.01 → critical 6.93.

**Step 8:** 20.014 > 6.93 → Reject H0. At the 0.01 level, there is sufficient evidence that mice are sensitive to door colour (at least one mean response time differs).

**TRAPS — where students lose marks**

$s^2_{\bar{x}}$ is the variance of the MEANS LIST {10.8, 22.2, 8.0} — put the three means into STAT mode and read sx, then square. It is NOT the average of the group variances.

Multiply by n (group size) exactly once: Between = n·$s^2_{\bar{x}}$.

Denominator df is k(n − 1) — with k = 3, n = 5: 3 × 4 = 12 (not 14, not 15).

If the table GIVES means and sd's (2018/19, 2017/18): jump straight to Step 3 — half the work is done.

F is always right-tailed here — one critical value.

A big visible difference can still fail: see the flights example (F = 2.316 < 3.47 → fail to reject!). Trust the F, not your eyes.

## PRACTICE · LEVEL 1 (warm-up, real 2016/17 Q7 = 2018/19 Q5: memory drug)

Memory drug: $\bar{x}$ = 83.4; Placebo: $\bar{x}$ = 50; No treatment: $\bar{x}$ = 16.6. All n = 5; variances 112.3, 109.0, 112.3. Test at α = 0.01.

**MODEL ANSWER — Level 1**

$$H_0: \mu_1 = \mu_2 = \mu_3, H_1: \text{ at least one differs}$$

Means given → jump to Step 3 directly.

$\bar{\bar{x}}$ = 50.0; $s_{\bar{x}}$ = sd{83.4, 50, 16.6} = 33.4 → $s^2_{\bar{x}}$ = 1115.56. Between = 5 × 1115.56 = 5577.8.

Within = (112.3 + 109.0 + 112.3)/3 = 111.2.

$$F = \frac{5577.8}{111.2} = 50.160$$

df = (2, 12), critical at 0.01 = 6.93. 50.160 > 6.93 → Reject H0: the treatments have different effects on exam scores.

## PRACTICE · LEVEL 2 (real exam: 2023/24 Q6 — flights, sd's given)

Delay times (min): AB103: −2, −1, −2, 2, −2, 0, −2, −3 (s = 1.58) • AB112: 19, −4, −5, −1, −4, 73, 0, 1 (s = 26.64) • AB108: 18, 60, 142, −1, −11, −1, 47, 13 (s = 50.25). (a) means, variance between, variance within; (b) test at α = 0.05 that the flights share the same mean delay.

### MODEL ANSWER — Level 2

Means: $\bar{x}1 = -10/8 = -1.250$; $\bar{x}2 = 79/8 = 9.875$; $\bar{x}3 = 267/8 = 33.375$. k = 3, n = 8.

$\bar{\bar{x}} = 14.000$; $s_{\bar{x}} = sd\{-1.25, 9.875, 33.375\} = 17.677 \rightarrow s^2_{\bar{x}} = 312.484$. Between = 8 × 312.484 = 2499.875.

$$s_p^2 = \frac{1.58^2 + 26.64^2 + 50.25^2}{3} = \frac{3237.249}{3} = 1079.083$$

$$F = \frac{2499.875}{1079.083} = 2.317$$

df = (2, 21), critical F(0.05) = 3.47. Since 2.316 < 3.47 → FAIL to reject H0.

At the 0.05 level there is not enough evidence that the mean delays differ. (The means LOOK wildly different — −1.25 vs 33.4 — but the within-group noise is huge. This is the newest paper's favourite trap.)

## PRACTICE · LEVEL 3 (monster: 2021/22 Q6 — full 20 marks, k = 4)

Training methods, 6 employees each. M1: 45, 59, 48, 46, 38, 47 • M2: 21, 12, 14, 17, 13, 17 • M3: 37, 32, 15, 25, 39, 41 • M4: 16, 11, 20, 21, 14, 7. Full ANOVA at α = 0.05.

### MODEL ANSWER — Level 3

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4, H_1: \ at \ \ least \ \ one \ \ differs$$

k = 4, n = 6.

STAT mode per group: $\bar{x}1 = 47.167$ ($s^2 = 46.167$), $\bar{x}2 = 15.667$ ($s^2 = 11.067$), $\bar{x}3 = 31.500$ ($s^2 = 98.300$), $\bar{x}4 = 14.833$ ($s^2 = 28.567$).

$\bar{\bar{x}} = 27.292$; $s_{\bar{x}} = sd\{47.167, 15.667, 31.5, 14.833\} = 15.309 \rightarrow s^2_{\bar{x}} = 234.359$. Between = 6 × 234.359 = 1406.153.

Within = (46.167 + 11.067 + 98.300 + 28.567)/4 = 46.025.

$$F = \frac{1406.153}{46.025} = 30.552$$

df = (3, 20), critical 3.10. 30.552 > 3.10 → Reject H0: the training methods differ significantly in performance.

**Monster factor:** four groups of raw data — pure endurance. Budget ~25 minutes and let the calculator's STAT mode do the eight means/sd's.

# PART 10 — THE MOCK FINAL (sit this Thursday 8:30–11:00 AM)

**Fresh numbers, exact real-paper format: 6 questions, 100 marks, 2.5 hours. Rules: answer booklet style, ALL calculations to 3 decimal places, no peeking at the key. Mark yourself with the answer key that follows; anything below full marks on a question → re-read that lesson's TEMPLATE box.**

## QUESTION 1 [10 MARKS]

a) A cafe owner records daily sales (RM ×10) for 12 random days: 42, 38, 45, 41, 39, 44, 40, 43, 46, 37, 42, 41. (i) Give a point estimate of the population mean. (ii) Construct and interpret a 95% confidence interval for μ. (7 marks)

b) A courier claims its mean delivery weight is more than 250 g. A sample of 40 parcels has mean 258 g; σ = 20 g. Test the claim at α = 0.05. (3 marks)

## QUESTION 2 [20 MARKS]

a) Two factories produce lamps. Factory 1: n = 45, $\bar{x}$ = 6.8 months lifespan, s = 1.1. Factory 2: n = 55, $\bar{x}$ = 7.3, s = 0.9. Assuming unequal variances, test at α = 0.05 whether the mean lifespans differ. State the hypotheses, the test statistic, the degrees of freedom, and the decision. (10 marks)

b) Six students take a speed-reading course. Scores before: 65, 70, 58, 62, 74, 68. After: 72, 78, 66, 69, 80, 71. At α = 0.05, does the course improve scores? (10 marks)

## QUESTION 3 [20 MARKS]

a) A website records visits: Mon 48, Tue 52, Wed 63, Thu 45, Fri 42. Test at α = 0.05 whether visits are equally distributed across the five days. (8 marks)

b) A survey classifies 250 customers by device (phone, laptop, tablet) and satisfaction: Satisfied — 45, 30, 15; Not satisfied — 45, 60, 55. At α = 0.05, is satisfaction related to device? (12 marks)

## QUESTION 4 [15 MARKS]

Hours studied x: 2, 4, 5, 7, 8, 10, 12. Exam score y: 65, 68, 72, 75, 80, 86, 90. a) Sketch the scatter plot — what relationship do you observe? (2 marks) b) Compute the Pearson correlation coefficient and interpret it. (11 marks) c) Is the correlation consistent with your sketch? (2 marks)

## QUESTION 5 [15 MARKS]

Using the same data as Question 4: a) Find $\Sigma x$, $\Sigma y$, $\Sigma xy$, $\Sigma x^2$, $\Sigma y^2$. (5 marks) b) Find the least-squares regression line. (6 marks) c) Predict the score of a student who studies 9 hours. (2 marks) d) Your friend studies 30 hours per week. Should you use this line to predict their score? Why? (2 marks)

## QUESTION 6 [20 MARKS]

Three delivery companies are timed (minutes) on the same route, 5 runs each. A: 12, 15, 14, 11, 13. B: 18, 20, 17, 22, 19. C: 9, 8, 11, 10, 12. a) Find the mean and standard deviation of each company. (6 marks) b) Compute the F statistic. (10 marks) c) At α = 0.05, do the companies differ in mean delivery time? (4 marks)

# MOCK FINAL — ANSWER KEY

### MODEL ANSWER — Q1a (7)

(i) $\bar{x}$ = 498/12 = 41.500.

(ii) s = 2.747, σ unknown, n = 12 → t, df = 11, 95% → column 0.025 → 2.201.

$$ME = 2.201 \times \frac{2.747}{\sqrt{12}} = 1.745, \qquad CI = (39.755,\ 43.245)$$

We are 95% confident the true mean daily sales lie between 39.755 and 43.245.

### MODEL ANSWER — Q1b (3)

$$H_0: \mu = 250, H_1: \mu > 250$$

$$z = \frac{258 - 250}{20/\sqrt{40}} = \frac{8}{3.162} = 2.530$$

Right-tailed critical 1.645. Since 2.530 > 1.645 → Reject H0 — sufficient evidence that mean weight exceeds 250 g.

### MODEL ANSWER — Q2a (10)

$$H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2 (two\text{-}tailed)$$

Unequal variances → Welch.

$$T = \frac{6.8 - 7.3}{\sqrt{\frac{1.1^2}{45} + \frac{0.9^2}{55}}} = \frac{-0.5}{0.204} = -2.451$$

$$v = \frac{(0.0269 + 0.0147)^2}{\frac{0.0269^2}{44} + \frac{0.0147^2}{54}} = 84.7 \quad \rightarrow \quad df = 84$$

Critical ±t(0.025, 84) ≈ ±1.989. −2.451 < −1.989 → Reject H0: the mean lifespans differ.

### MODEL ANSWER — Q2b (10)

Paired. D = after − before = 7, 8, 8, 7, 6, 3. $\bar{D}$ = 39/6 = 6.500, sD = 1.871.

$$H_0: \mu_D = 0, H_1: \mu_D > 0$$

$$t = \frac{6.500}{1.871/\sqrt{6}} = \frac{6.500}{0.764} = 8.510$$

df = 5, critical t(0.05,5) = 2.015. Reject H0 — the course significantly improves scores.

### MODEL ANSWER — Q3a (8)

H0: visits equal across days; H1: at least one differs. Total 250 → e = 50 each.

$$\chi^2 = \frac{4 + 4 + 169 + 25 + 64}{50} = \frac{266}{50} = 5.320$$

df = 4 → critical 9.488. 5.320 < 9.488 → Fail to reject H0: no evidence the visits are unequal.

### MODEL ANSWER — Q3b (12)

H0: satisfaction independent of device; H1: related. Row totals 90, 160; column totals 90, 90, 70; n = 250.

Expected: satisfied 32.4, 32.4, 25.2; not satisfied 57.6, 57.6, 44.8.

$$\chi^2 = \frac{12.6^2}{32.4} + \frac{2.4^2}{32.4} + \frac{10.2^2}{25.2} + \frac{12.6^2}{57.6} + \frac{2.4^2}{57.6} + \frac{10.2^2}{44.8}$$

$$= 4.900 + 0.178 + 4.129 + 2.756 + 0.100 + 2.322 = 14.385$$

df = (2−1)(3−1) = 2 → critical 5.991. 14.385 > 5.991 → Reject H0: satisfaction IS related to device type.

## MODEL ANSWER — Q4 (15)

(a) Points rise together → positive linear relationship.

(b) Σx = 48, Σy = 536, Σxy = 3867, Σx² = 402, Σy² = 41554, n = 7.

$$numerator = 3867 - \frac{48 \times 536}{7} = 191.571, \quad denominator = \sqrt{72.857 \times 512.857} =$$

$$r = \frac{191.571}{193.086} = 0.992$$

Very strong positive linear relationship: more study hours, higher scores. (c) Yes — consistent with the rising scatter.

## MODEL ANSWER — Q5 (15)

(a) Same sums as Q4 (compute once, use twice!).

$$b_1 = \frac{191.571}{72.857} = 2.629, b_0 = 76.571 - 2.629(6.857) = 58.541, \quad \hat{y} = 58.541 + 2.62$$

(c) ŷ(9) = 58.541 + 2.629(9) = 82.206 ≈ 82.2 marks.

(d) No — 30 hours is far outside the data range (2–12). Extrapolation is unreliable.

## MODEL ANSWER — Q6 (20)

(a) A: $\bar{x}$ = 13.0, s = 1.581 ($s^2$ = 2.5). B: $\bar{x}$ = 19.2, s = 1.924 ($s^2$ = 3.7). C: $\bar{x}$ = 10.0, s = 1.581 ($s^2$ = 2.5).

(b) $\bar{\bar{x}}$ = 14.067; $s_{\bar{x}}$ = sd{13, 19.2, 10} = 4.692 → $s^2_{\bar{x}}$ = 22.013. Between = 5 × 22.013 = 110.067. Within = (2.5 + 3.7 + 2.5)/3 = 2.900.

$$F = \frac{110.067}{2.900} = 37.954$$

(c)

$$H_0: \mu_A = \mu_B = \mu_C, H_1: \text{ at least one differs}$$

df = (2, 12) → critical 3.89. 37.954 > 3.89 → Reject H0: the companies differ in mean delivery time.

**Scoring guide:** 80+ → you are ready, spend the rest of the morning on the Most Wanted list. 60–79 → redo the TEMPLATE boxes of the questions you lost. Below 60 → re-run Lessons 4, 5, 6, 9 worked examples with the answers covered.

# PART 11 — EXAM-DAY PROTOCOL (Thursday 16 July, 2:30 PM)

## The last-hour review (do at 12:00–12:45)

Read ONLY these, in order: the Chooser table (0.6) → each lesson's TEMPLATE box → each lesson's TRAPS box → the four z numbers (1.645, 1.960, 2.326, 2.576). Nothing else. New material in the last hour hurts more than it helps.

## In the hall

- Bring: IC/matric card, TWO pens, calculator (charged/spare battery), water. Arrive 2:00 PM.
- First 3 minutes: skim all 6 questions. Mark the ones that look like your strongest lessons; do those first. Likely order of ease for you: Q1 → Q6 → Q3 → Q2 → Q5 → Q4 (adjust to taste).
- Budget ≈ 1.5 min per mark: Q1 ~15 min, 20-markers ~30 min. Leave 15 minutes to re-check decisions and 3-decimal rounding.
- Every hypothesis test: write ALL 5 steps even if the arithmetic scares you — steps carry most of the marks.
- Stuck on arithmetic? Write the formula with numbers substituted and move on. Substitution alone earns method marks.
- Draw and shade the little bell curve in every test — free marks, every time.
- Before submitting: check every conclusion sentence answers the question's OWN words ('the company is cheating', 'mice are sensitive to colour' …).

## Formula sheet map (printed in your paper — know where things live)

| Section on sheet | Used in | Lesson |
|---|---|---|
| (a) CI / point estimate $\hat{p} \pm z\sqrt{(\hat{p}\hat{q}/n)}$ | Q1 | L1 |
| (b) One-sample: z, t, $\chi^2 = (n-1)s^2/\sigma^2$ | Q1/Q2 | L2, L3 |
| (c) Two samples: z, pooled t + $sp^2$, Welch T + v, F = $s1^2/s2^2$, paired $\bar{D}$, sD | Q2 | L4, L5 |
| (d) Chi-square $\Sigma(o-e)^2/e$; e = row×col/total; df rules | Q3 | L6 |
| (e) Correlation: Pearson r, Spearman rs, $t = r/\sqrt{[(1-r^2)/(n-2)]}$ | Q4 | L7 |
| (f) Regression: b1, b0; slope t = (b1−β1)/sb1 | Q5 | L8 |
| (g) ANOVA: $F = n \cdot s^2\bar{x}/s^2p$; df = k−1, k(n−1) | Q6 | L9 |

## Glossary — English → Arabic (the exam's scary words)

| English | Arabic | English | Arabic |
|---|---|---|---|
| population / sample | المجتمع / العينة | significance level α | مستوى الدلالة |
| mean / average | المتوسط | critical value | القيمة الحرجة |
| standard deviation | الانحراف المعياري | rejection region | منطقة الرفض |
| variance | التباين | reject / fail to reject | نرفض / لا نرفض |
| proportion | النسبة | sufficient evidence | دليل كافٍ |
| point estimate | التقدير النقطي | claim | الادعاء |

| | | | |
|---|---|---|---|
| confidence interval | فترة الثقة | test statistic | إحصائية الاختبار |
| margin of error | هامش الخطأ | degrees of freedom | درجات الحرية |
| hypothesis testing | اختبار الفرضيات | goodness-of-fit | اختبار حسن المطابقة |
| null / alternative hypothesis | فرضية العدم / البديلة | independence / related | الاستقلالية / مرتبط |
| two-tailed / one-tailed | ذيلان / ذيل واحد | observed / expected | المشاهَد / المتوقع |
| paired samples | عينات مزدوجة (قبل/بعد) | correlation coefficient | معامل الارتباط |
| pooled variance | التباين المجمّع | regression line / slope / intercept | خط الانحدار / الميل / التقاطع |
| normally distributed | يتبع التوزيع الطبيعي | predict / estimate | يتنبأ / يقدّر |
| random sample | عينة عشوائية | at least one mean differs | متوسط واحد على الأقل مختلف |

**Final word.** Eleven years of papers, one pattern: the same nine lessons, the same five steps, numbers changed. You have now seen every costume this exam owns — including the exact questions they recycle. Work the schedule, sit the mock, sleep well, pray, and walk in knowing the paper was written for the student you will be by Thursday. Bi tawfeeq — بالتوفيق!